Appendix D: R Code

Melat Tarekegn

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Get working directory and read the data

```
getwd()
```

```
## [1] "/Users/melattarekegn/Downloads/DATA"
```

```
data <- read.csv("Mental Health Care.csv", sep=",")
library(ggplot2)
library(stringr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# Descriptive Statistics
nrow(data)
```

```
## [1] 10404
```

```
ncol(data)
```

```
## [1] 15
```

```
names(data)
```

```
##  [1] "Indicator"              "Group"                  "State"
##  [4] "Subgroup"               "Phase"                  "Time.Period"
##  [7] "Time.Period.Label"      "Time.Period.Start.Date" "Time.Period.End.Date"
## [10] "Value"                  "LowCI"                  "HighCI"
## [13] "Confidence.Interval"    "Quartile.Range"         "Suppression.Flag"
```

```r
head(data)
```

```
##                                                      Indicator
## 1 Took Prescription Medication for Mental Health, Last 4 Weeks
## 2 Took Prescription Medication for Mental Health, Last 4 Weeks
## 3 Took Prescription Medication for Mental Health, Last 4 Weeks
## 4 Took Prescription Medication for Mental Health, Last 4 Weeks
## 5 Took Prescription Medication for Mental Health, Last 4 Weeks
## 6 Took Prescription Medication for Mental Health, Last 4 Weeks
##               Group         State      Subgroup Phase Time.Period
## 1 National Estimate United States United States     2          13
## 2            By Age United States 18 - 29 years     2          13
## 3            By Age United States 30 - 39 years     2          13
## 4            By Age United States 40 - 49 years     2          13
## 5            By Age United States 50 - 59 years     2          13
## 6            By Age United States 60 - 69 years     2          13
##       Time.Period.Label Time.Period.Start.Date Time.Period.End.Date Value LowCI
## 1 Aug 19 - Aug 31, 2020             08/19/2020           08/31/2020  19.4  19.0
## 2 Aug 19 - Aug 31, 2020             08/19/2020           08/31/2020  18.7  17.2
## 3 Aug 19 - Aug 31, 2020             08/19/2020           08/31/2020  18.3  17.3
## 4 Aug 19 - Aug 31, 2020             08/19/2020           08/31/2020  20.4  19.5
## 5 Aug 19 - Aug 31, 2020             08/19/2020           08/31/2020  21.2  20.2
## 6 Aug 19 - Aug 31, 2020             08/19/2020           08/31/2020  19.7  18.7
##   HighCI Confidence.Interval Quartile.Range Suppression.Flag
## 1   19.8         19.0 - 19.8                              NA
## 2   20.3         17.2 - 20.3                              NA
## 3   19.2         17.3 - 19.2                              NA
## 4   21.3         19.5 - 21.3                              NA
## 5   22.2         20.2 - 22.2                              NA
## 6   20.7         18.7 - 20.7                              NA
```

```r
tail(data)
```

```
##                                                          Indicator    Group
## 10399 Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks By State
## 10400 Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks By State
## 10401 Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks By State
## 10402 Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks By State
## 10403 Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks By State
## 10404 Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks By State
##               State      Subgroup Phase Time.Period    Time.Period.Label
## 10399       Vermont       Vermont   3.4          45 Apr 27 - May 9, 2022
## 10400      Virginia      Virginia   3.4          45 Apr 27 - May 9, 2022
## 10401    Washington    Washington   3.4          45 Apr 27 - May 9, 2022
## 10402 West Virginia West Virginia   3.4          45 Apr 27 - May 9, 2022
## 10403     Wisconsin     Wisconsin   3.4          45 Apr 27 - May 9, 2022
## 10404       Wyoming       Wyoming   3.4          45 Apr 27 - May 9, 2022
##       Time.Period.Start.Date Time.Period.End.Date Value LowCI HighCI
## 10399             04/27/2022           05/09/2022  11.9   8.2   16.5
## 10400             04/27/2022           05/09/2022  10.1   7.1   13.8
## 10401             04/27/2022           05/09/2022  14.8  12.5   17.4
## 10402             04/27/2022           05/09/2022   9.8   6.6   13.8
## 10403             04/27/2022           05/09/2022  11.6   8.5   15.3
```

```
## 10404             04/27/2022           05/09/2022  11.5   8.6   15.1
##       Confidence.Interval Quartile.Range Suppression.Flag
## 10399          8.2 - 16.5      11.4-12.9               NA
## 10400          7.1 - 13.8       9.6-11.3               NA
## 10401         12.5 - 17.4      13.0-20.8               NA
## 10402          6.6 - 13.8       9.6-11.3               NA
## 10403          8.5 - 15.3      11.4-12.9               NA
## 10404          8.6 - 15.1      11.4-12.9               NA
```

```r
summary(data)
```

```
##   Indicator            Group              State              Subgroup
##  Length:10404       Length:10404       Length:10404       Length:10404
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##     Phase             Time.Period    Time.Period.Label  Time.Period.Start.Date
##  Length:10404       Min.   : 1.00   Length:10404       Length:10404
##  Class :character   1st Qu.:20.00   Class :character   Class :character
##  Mode  :character   Median :29.00   Mode  :character   Mode  :character
##                     Mean   :28.13
##                     3rd Qu.:37.00
##                     Max.   :45.00
##
##  Time.Period.End.Date     Value           LowCI           HighCI
##  Length:10404         Min.   : 1.40   Min.   : 0.80   Min.   : 2.00
##  Class :character     1st Qu.:10.30   1st Qu.: 8.00   1st Qu.:12.90
##  Mode  :character     Median :16.20   Median :13.90   Median :19.20
##                       Mean   :17.45   Mean   :14.77   Mean   :20.48
##                       3rd Qu.:24.00   3rd Qu.:20.80   3rd Qu.:27.40
##                       Max.   :62.90   Max.   :53.20   Max.   :71.90
##                       NA's   :490     NA's   :490     NA's   :490
##  Confidence.Interval Quartile.Range     Suppression.Flag
##  Length:10404        Length:10404       Min.   :1
##  Class :character    Class :character   1st Qu.:1
##  Mode  :character    Mode  :character   Median :1
##                                         Mean   :1
##                                         3rd Qu.:1
##                                         Max.   :1
##                                         NA's   :10382
```

```r
colSums(is.na(data)) # Check for missing values
```

```
##              Indicator                  Group                  State
##                      0                      0                      0
##               Subgroup                  Phase            Time.Period
##                      0                      0                      0
##      Time.Period.Label Time.Period.Start.Date   Time.Period.End.Date
##                      0                      0                      0
##                  Value                  LowCI                 HighCI
```

```
##                     490                     490                     490
##    Confidence.Interval         Quartile.Range       Suppression.Flag
##                      0                       0                  10382
```

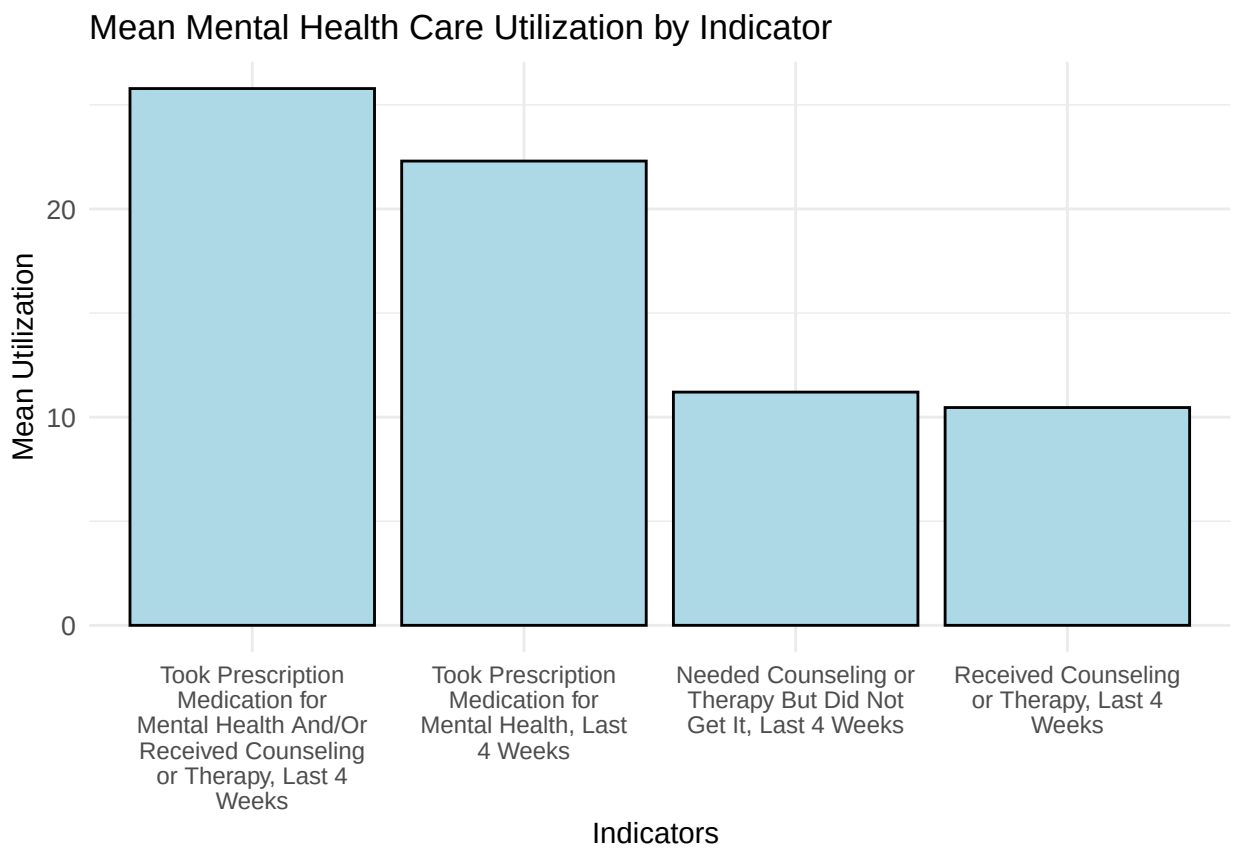
Summarize mental health care utilization by Indicator and view the summary

```
Indicatorsummary <- data %>%
  group_by(Indicator) %>%
  summarize(
    mean_utilization = mean(Value, na.rm = TRUE),
    median_utilization = median(Value, na.rm = TRUE),
    sd_utilization = sd(Value, na.rm = TRUE))
print(Indicatorsummary)
```

```
## # A tibble: 4 x 4
##   Indicator                   mean_utilization median_utilization sd_utilization
##   <chr>                                  <dbl>              <dbl>          <dbl>
## 1 Needed Counseling or Thera~             11.2               10.7           4.41
## 2 Received Counseling or The~             10.5               10             3.83
## 3 Took Prescription Medicati~             25.8               25.7           5.63
## 4 Took Prescription Medicati~             22.3               22.1           5.18
```

Visualize Indicator summary as a bar chart. Use actual mean_utilization values (rather than counting occurrences(stat = "identity")).

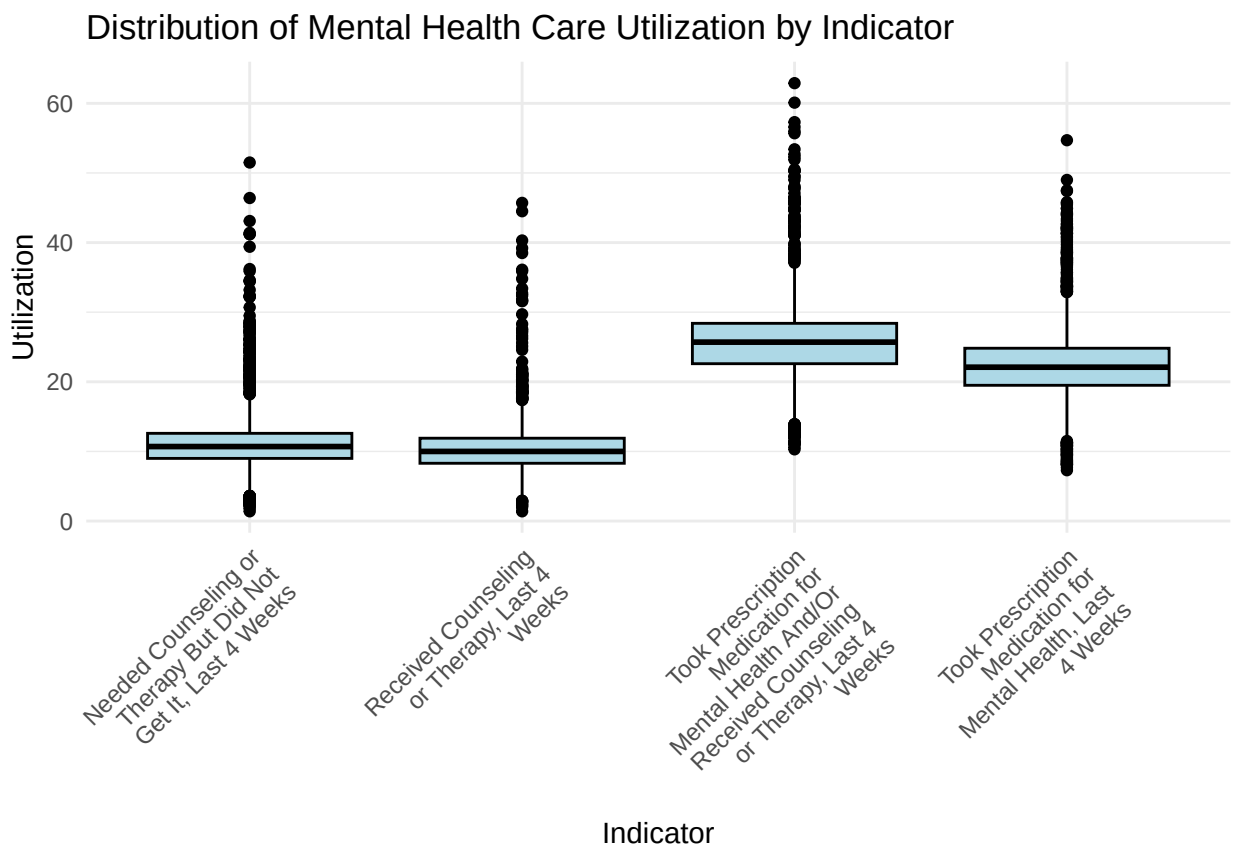
```
ggplot(Indicatorsummary, aes(x = reorder(Indicator, -mean_utilization), y = mean_utilization)) +
  geom_bar(stat = "identity", fill = "lightblue", color="black") +
  labs(title = "Mean Mental Health Care Utilization by Indicator",
       x = "Indicators ",
       y = "Mean Utilization") +
  theme_minimal() + theme(legend.position = "none",  # Hide legend if not needed
                       axis.text.y = element_text(size = 10)) +  # Adjust font size
  scale_y_continuous(labels = scales::comma) +  # Format y-axis numbers
  scale_x_discrete(labels = function(x) str_wrap(x, width = 20))  # Wrap text
```



Create a box plot to visualize the spread of mental health care utilization across indicators

```
ggplot(data, aes(x = Indicator, y = Value)) +
  geom_boxplot(fill = "lightblue", color = "black") +
  labs(title = "Distribution of Mental Health Care Utilization by Indicator",
       x = "Indicator",
       y = "Utilization") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  # Rotate x-axis labels
  scale_y_continuous(labels = scales::comma) +  # Format y-axis numbers
  scale_x_discrete(labels = function(x) str_wrap(x, width = 20))  # Wrap text for readability
```

```
## Warning: Removed 490 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Summarize mental health care utilization by state and view the summary

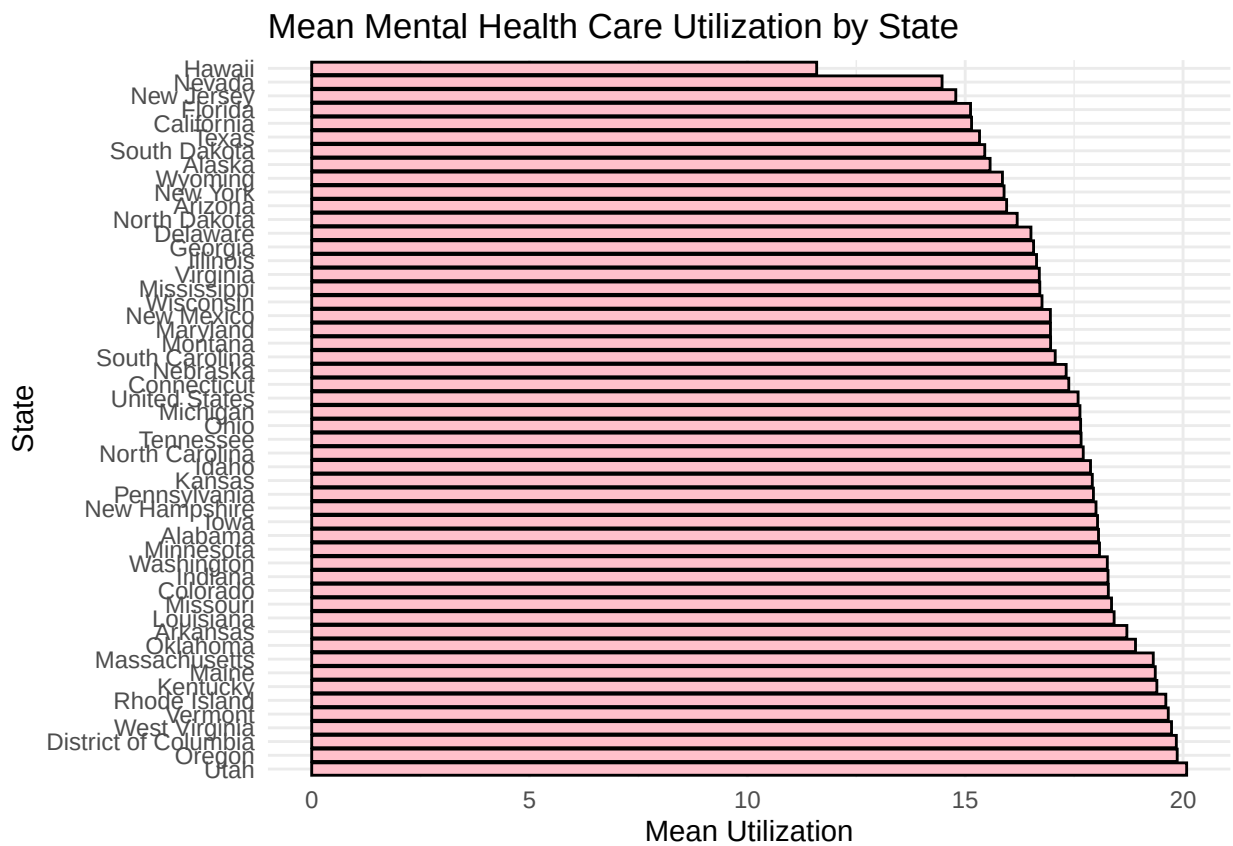
```
state_summary <- data %>%
  group_by(State) %>%
  summarize(
    mean_utilization = mean(Value, na.rm = TRUE),
    median_utilization = median(Value, na.rm = TRUE),
    sd_utilization = sd(Value, na.rm = TRUE))
print(state_summary)
```

```
## # A tibble: 52 x 4
##    State                mean_utilization median_utilization sd_utilization
##    <chr>                           <dbl>              <dbl>          <dbl>
##  1 Alabama                          18.1               17             8.38
##  2 Alaska                           15.6               14.9           5.34
##  3 Arizona                          16.0               16.5           5.93
##  4 Arkansas                         18.7               17.7           9.43
##  5 California                       15.1               14.6           4.48
##  6 Colorado                         18.3               17.0           6.32
##  7 Connecticut                      17.4               17.0           6.94
##  8 Delaware                         16.5               16             7.16
##  9 District of Columbia             19.8               19.5           5.83
## 10 Florida                          15.1               14.6           6.20
## # i 42 more rows
```

Visualize State summary as a column chart. It also reorders states so that those with the highest

mean_utilization appear at the top.Use actual mean_utilization values (rather than counting occurrences(stat = "identity")).

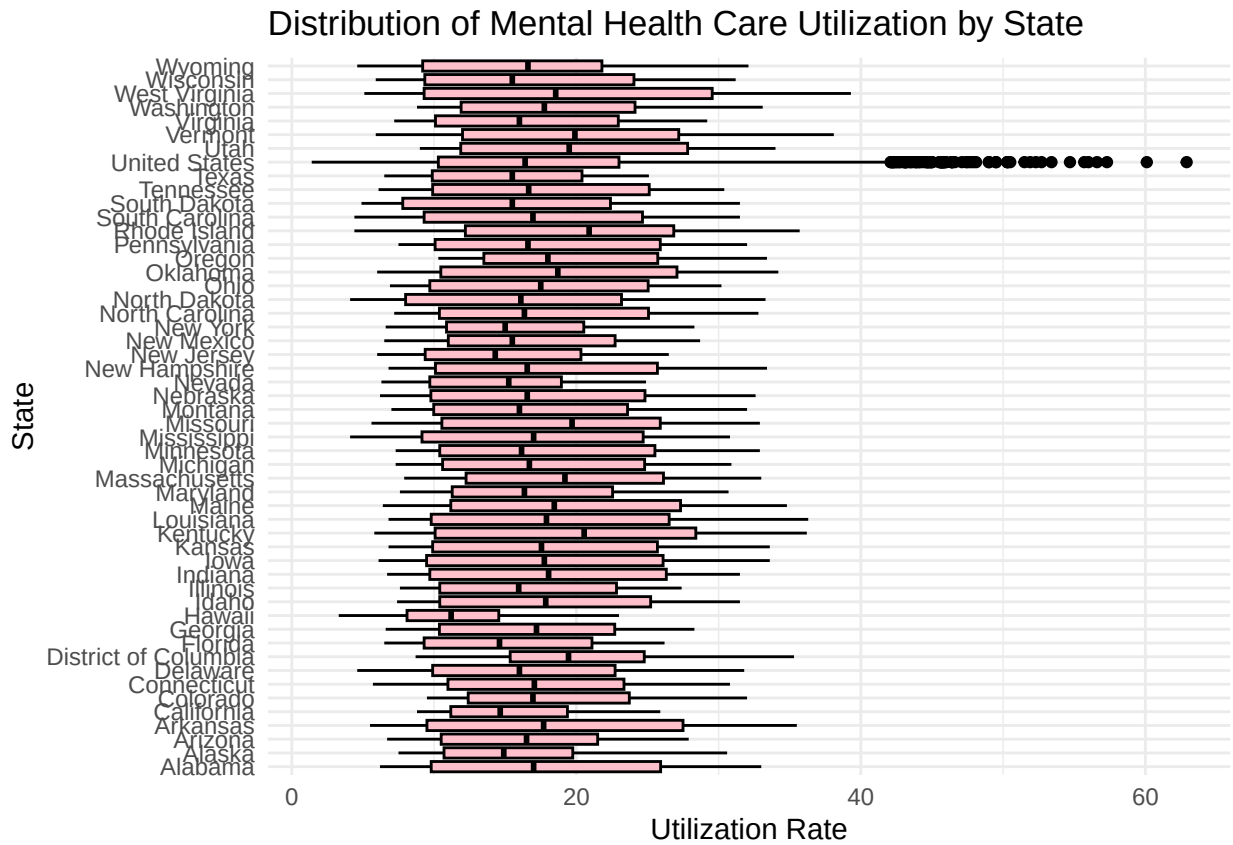
```
ggplot(state_summary, aes(x = reorder(State, -mean_utilization), y = mean_utilization)) +
  geom_bar(stat = "identity", fill = "pink", color="black") +
  coord_flip() +
  labs(title = "Mean Mental Health Care Utilization by State",
       x = "State",
       y = "Mean Utilization") +
  theme_minimal()
```



Create a box plot that shows the distribution / spread and variation of mental health care utilization across the different states

```
ggplot(data, aes(x = State, y = Value)) +
  geom_boxplot(fill = "pink", color = "black") +
  coord_flip() +
  labs(title = "Distribution of Mental Health Care Utilization by State",
       x = "State",
       y = "Utilization Rate") +
  theme_minimal()
```

```
## Warning: Removed 490 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Regression analysis, correlation analysis and hypothesis testing

```
# ================= REGRESSION ANALYSIS ================= #
# Convert categorical state variable to numeric factor for regression
state_summary$StateNumeric <- as.numeric(as.factor(state_summary$State))

# Perform linear regression
regression_model1 <- lm(mean_utilization ~ StateNumeric, data = state_summary)
summary(regression_model1)
```

```
##
## Call:
## lm(formula = mean_utilization ~ StateNumeric, data = state_summary)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.5580 -1.0301  0.1523  1.1074  2.7467
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.95163    0.46633  36.351   <2e-16 ***
## StateNumeric  0.01644    0.01531   1.073    0.288
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.657 on 50 degrees of freedom
```

```
## Multiple R-squared:  0.02252,    Adjusted R-squared:  0.002972
## F-statistic: 1.152 on 1 and 50 DF,  p-value: 0.2883
```

```r
# ================= CORRELATION ANALYSIS ================= #
# Spearman correlation test (since State is categorical)
correlation_test1 <- cor.test(state_summary$StateNumeric, state_summary$mean_utilization, method = "spea
print(correlation_test1)
```

```
##
##  Spearman's rank correlation rho
##
## data:  state_summary$StateNumeric and state_summary$mean_utilization
## S = 21098, p-value = 0.4822
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.09937676
```

```r
# ================= HYPOTHESIS TESTING ================= #
# Checking normality assumption
shapiro_test1 <- shapiro.test(state_summary$mean_utilization)  # If p-value < 0.05, data is NOT normal
print(shapiro_test1)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  state_summary$mean_utilization
## W = 0.9542, p-value = 0.04398
```

```r
# If data is NOT normal, use Kruskal-Wallis test instead of ANOVA
kruskal_test1 <- kruskal.test(mean_utilization ~ as.factor(State), data = state_summary)
print(kruskal_test1)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  mean_utilization by as.factor(State)
## Kruskal-Wallis chi-squared = 51, df = 51, p-value = 0.4737
```

Summarize average utilization by the different groups and view the summary

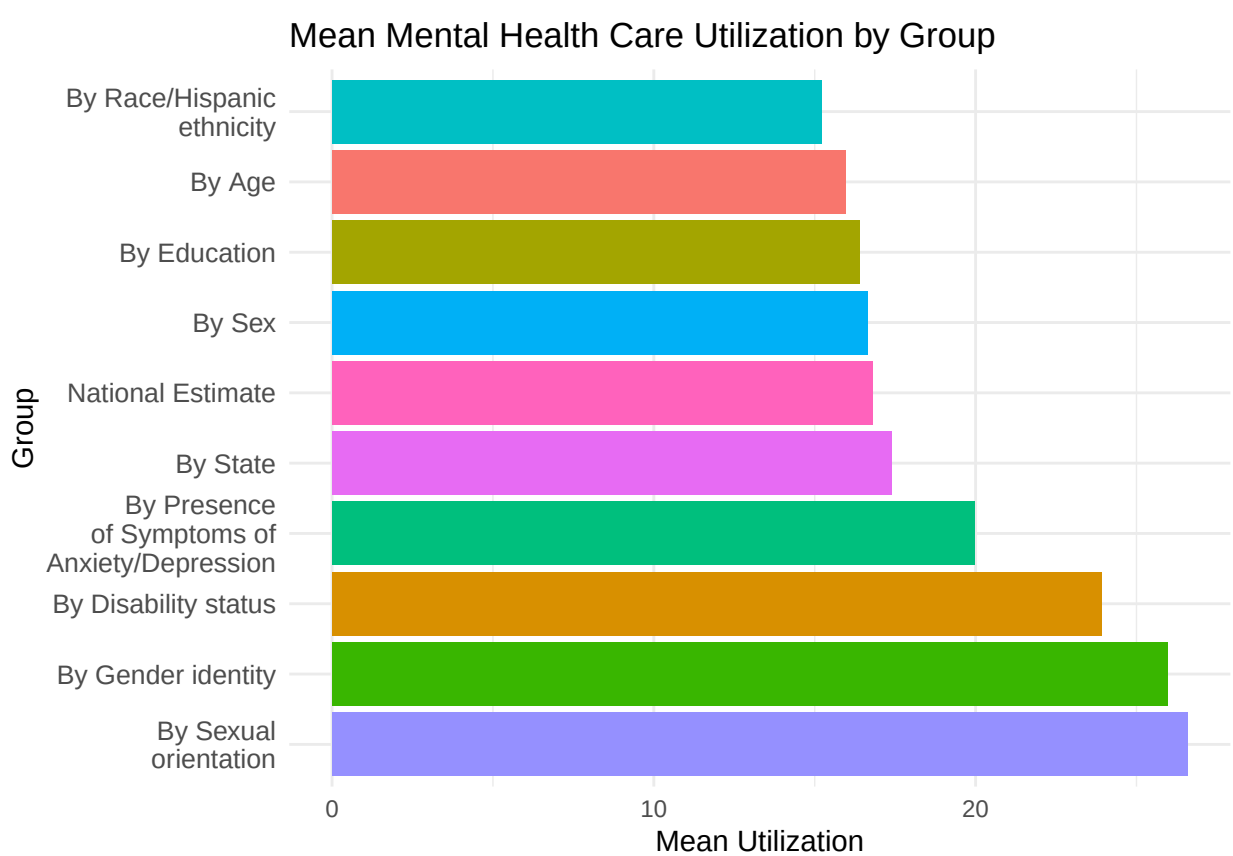
```r
group_summary <- data %>%
  group_by(Group) %>%
  summarize(mean_utilization = mean(Value, na.rm = TRUE),
            median_utilization = median(Value, na.rm = TRUE),
            sd_utilization = sd(Value, na.rm = TRUE))
print(group_summary)
```

```
## # A tibble: 10 x 4
##    Group                      mean_utilization median_utilization sd_utilization
##    <chr>                                 <dbl>              <dbl>          <dbl>
```

```
##  1 By Age                                16.0              16.9          7.72
##  2 By Disability status                  23.9              21.1         13.2
##  3 By Education                          16.4              15.0          6.70
##  4 By Gender identity                    26.0              23.2         15.2
##  5 By Presence of Symptoms o~            20.0              17.4         12.6
##  6 By Race/Hispanic ethnicity            15.2              13.4          6.72
##  7 By Sex                                16.6              14.4          7.81
##  8 By Sexual orientation                 26.6              24.8         11.3
##  9 By State                              17.4              16.1          7.66
## 10 National Estimate                     16.8              15.9          6.50
```

Visualize the mental health service utlization of the different groups

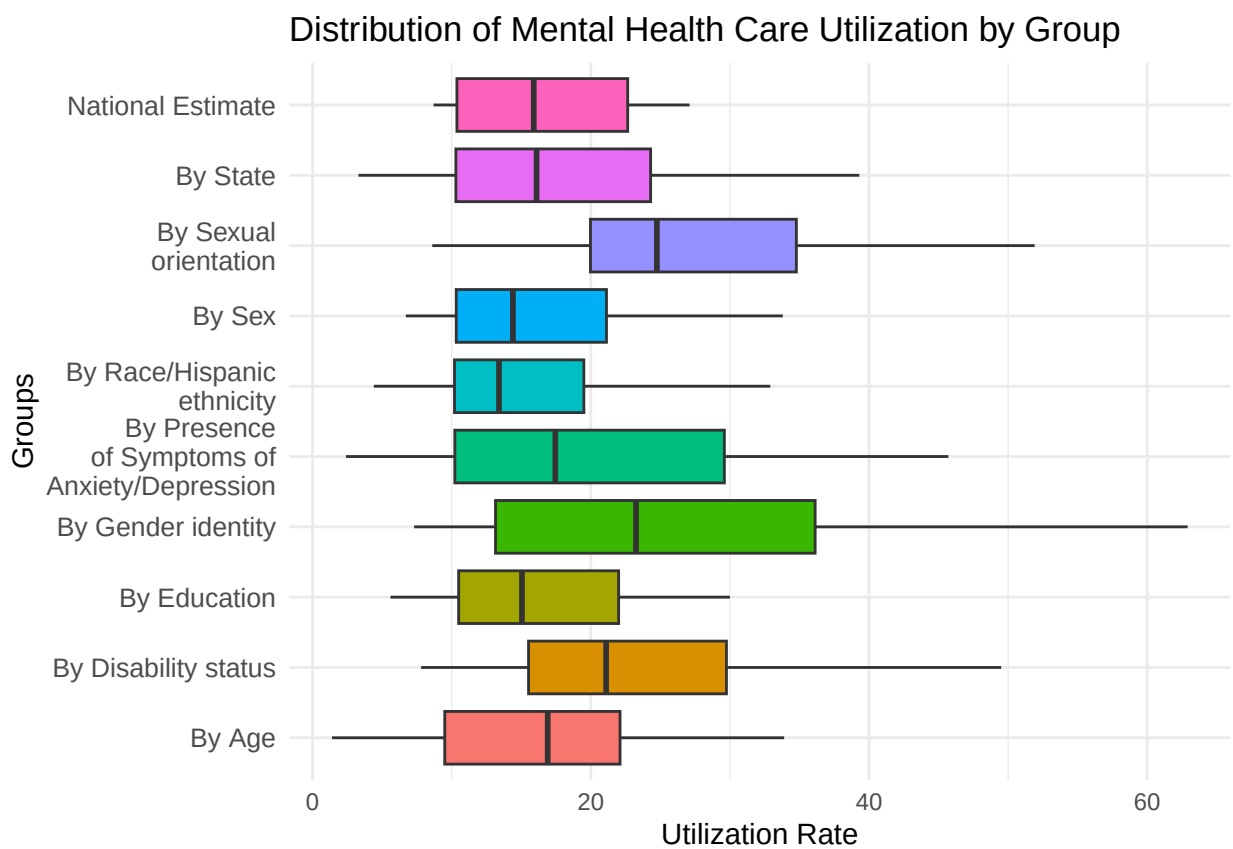
```
ggplot(group_summary, aes(x = reorder(Group, -mean_utilization), y = mean_utilization, fill = Group)) +
  geom_bar(stat = "identity") +
  coord_flip() +  # Flip for better readability
  labs(title = "Mean Mental Health Care Utilization by Group",
       x = "Group",
       y = "Mean Utilization") +
  theme_minimal() +
  theme(legend.position = "none",  # Hide legend if not needed
        axis.text.y = element_text(size = 10)) +  # Adjust font size
  scale_y_continuous(labels = scales::comma) +  # Format y-axis numbers
  scale_x_discrete(labels = function(x) str_wrap(x, width = 20))  # Wrap text
```



Create a box plot to see the distribution of utilization across different groups

```r
ggplot(data, aes(x = Group, y = Value, fill = Group)) +
  geom_boxplot() +
  labs(title = "Distribution of Mental Health Care Utilization by Group",
       x = "Groups",
       y = "Utilization Rate") +
  theme_minimal() +
  theme(legend.position = "none",  # Hide legend if not needed
        axis.text.y = element_text(size = 10)) +  # Adjust font size
  scale_y_continuous(labels = scales::comma) +  # Format y-axis numbers
  scale_x_discrete(labels = function(x) str_wrap(x, width = 20) #wrap the text
                   ) + coord_flip()  # Flip for readability
```

```
## Warning: Removed 490 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Hypothesis testing for data item group

```r
# Checking normality assumption
shapiro_test2 <- shapiro.test(group_summary$mean_utilization)  # If p-value < 0.05, data is NOT normal
print(shapiro_test2)
```

```
##
##  Shapiro-Wilk normality test
##
```

```
## data:  group_summary$mean_utilization
## W = 0.81746, p-value = 0.02361
```

```r
# If data is NOT normal, use Kruskal-Wallis test instead of ANOVA
kruskal_test2 <- kruskal.test(mean_utilization ~ as.factor(Group), data = group_summary)
print(kruskal_test2)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  mean_utilization by as.factor(Group)
## Kruskal-Wallis chi-squared = 9, df = 9, p-value = 0.4373
```

Summarize average utilization by the different subgroups and view the summary

```r
subgroup_summary <- data %>%
  group_by(Subgroup) %>%
  summarize(mean_utilization = mean(Value, na.rm = TRUE),
            median_utilization = median(Value, na.rm = TRUE),
            sd_utilization = sd(Value, na.rm = TRUE))
print(subgroup_summary)
```

```
## # A tibble: 80 x 4
##    Subgroup           mean_utilization median_utilization sd_utilization
##    <chr>                         <dbl>              <dbl>          <dbl>
##  1 18 - 29 years                  20.8               20.7           4.61
##  2 30 - 39 years                  19.7               17.8           5.28
##  3 40 - 49 years                  17.8               16.9           6.46
##  4 50 - 59 years                  16.6               15.6           7.67
##  5 60 - 69 years                  14.2               13.4           8.31
##  6 70 - 79 years                  11.8               11.0           8.17
##  7 80 years and above             10.3               13.4           6.40
##  8 Alabama                        18.1               17             8.38
##  9 Alaska                         15.6               14.9           5.34
## 10 Arizona                        16.0               16.5           5.93
## # i 70 more rows
```

Calculate mean utilization for each group and subgroup and sort by group

```r
groupsummary <- data %>%
  group_by(Group, Subgroup) %>%
  summarize(mean_utilization = mean(Value, na.rm = TRUE)) %>%
  arrange(Group, desc(mean_utilization))  # Sort by Group first, then by utilization
```

```
## `summarise()` has grouped output by 'Group'. You can override using the
## `.groups` argument.
```

```r
print(groupsummary)
```

```
## # A tibble: 80 x 3
## # Groups:   Group [10]
```

```
##    Group                Subgroup                        mean_utilization
##    <chr>                <chr>                                      <dbl>
##  1 By Age               18 - 29 years                               20.8
##  2 By Age               30 - 39 years                               19.7
##  3 By Age               40 - 49 years                               17.8
##  4 By Age               50 - 59 years                               16.6
##  5 By Age               60 - 69 years                               14.2
##  6 By Age               70 - 79 years                               11.8
##  7 By Age               80 years and above                          10.3
##  8 By Disability status With disability                             33.2
##  9 By Disability status Without disability                          14.6
## 10 By Education         Some college/Associate's degree             18.8
## # i 70 more rows
```

Remove the state summary since they are examined separately

```
top_subgroups <- groupsummary %>%
  filter(Group != "By State") %>%  # Remove states since they are examined alone
  group_by(Group) %>%
  top_n(10, mean_utilization)  # Keep only the top 10 subgroups per Group
print(top_subgroups)
```

```
## # A tibble: 29 x 3
## # Groups:   Group [9]
##    Group                Subgroup                        mean_utilization
##    <chr>                <chr>                                      <dbl>
##  1 By Age               18 - 29 years                               20.8
##  2 By Age               30 - 39 years                               19.7
##  3 By Age               40 - 49 years                               17.8
##  4 By Age               50 - 59 years                               16.6
##  5 By Age               60 - 69 years                               14.2
##  6 By Age               70 - 79 years                               11.8
##  7 By Age               80 years and above                          10.3
##  8 By Disability status With disability                             33.2
##  9 By Disability status Without disability                          14.6
## 10 By Education         Some college/Associate's degree             18.8
## # i 19 more rows
```

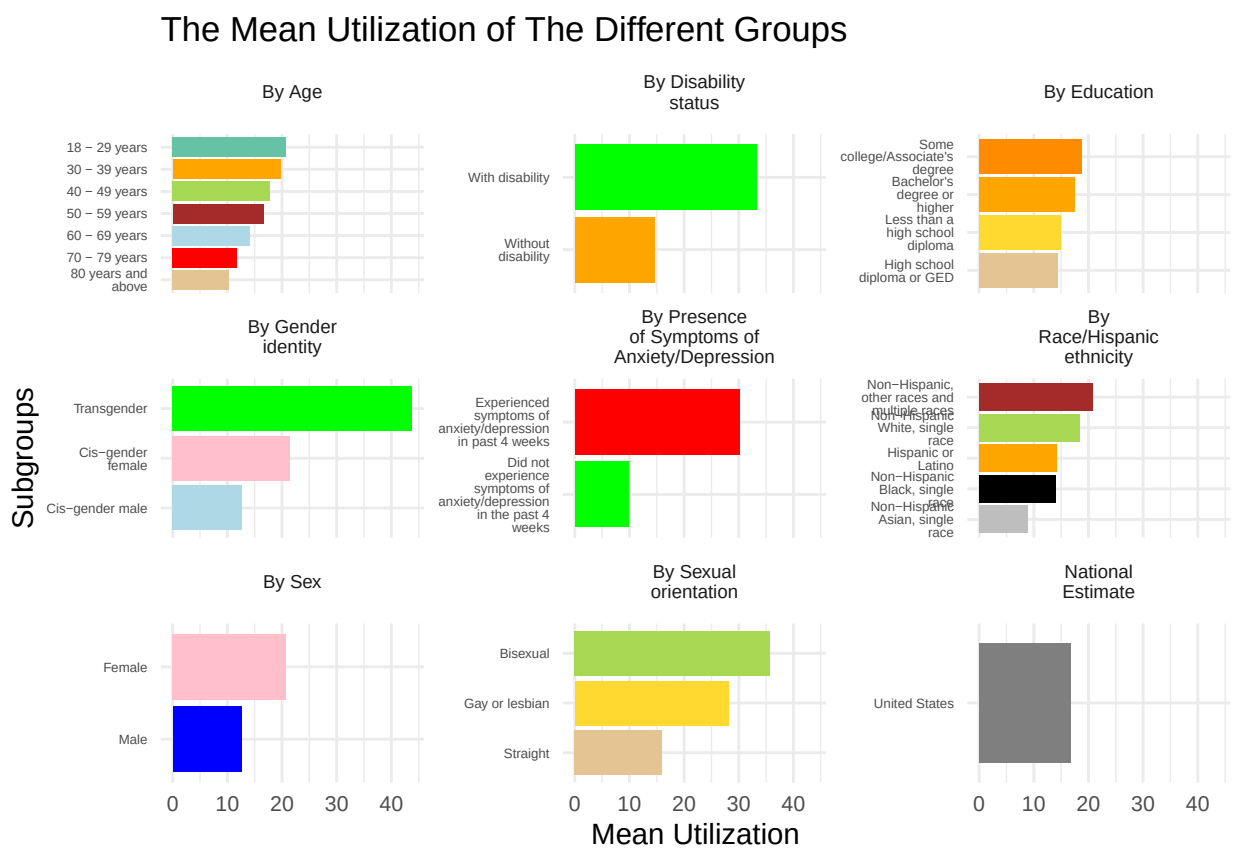
Select color for all the subgroups to be able to filter out the specific subgroups and plot them.

```
subgroup_colors <- c(
  "18 - 29 years" = "#66c2a5", "30 - 39 years" ="orange","40 - 49 years"="#a6d854", "50 - 59 years"="bro
  "Male" = "blue", "Female" = "pink", "With disability" ="green", "Without disability"="orange",
  "Some college/Associate's degree"="darkorange", "Bachelor's degree or higher"="orange", "Less than a h
  "Cis-gender female"="pink", "Cis-gender male"="lightblue", "Transgender"="green","Experienced symptoms
  "Non-Hispanic, other races and multiple races"="brown", "Non-Hispanic White, single race"="#a6d854","N
  "Bisexual" = "#a6d854","Gay or lesbian" = "#ffd92f","Straight" = "#e5c494")
```

Display the different subgroups arranged by the groups.

```
ggplot(top_subgroups, aes(x = reorder(Subgroup, mean_utilization), y = mean_utilization, fill = Subgroup
  geom_col() +
```

```
  facet_wrap(~ Group, scales = "free_y", labeller = labeller(Group = function(x) stringr::str_wrap(x, w
  theme_minimal() +
  labs(
    title = "The Mean Utilization of The Different Groups",
    x = "Subgroups",
    y = "Mean Utilization"
  ) +
  theme(
    axis.text.y = element_text(size = 4.5),
    axis.text.x = element_text(size = 8, vjust = 1),
    strip.text = element_text(size = 7, margin = margin(b = 5)),
    legend.position = "none"
  ) +
  scale_y_continuous(labels = scales::comma) +
  scale_fill_manual(values = subgroup_colors) +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 15)) +
  coord_flip()
```

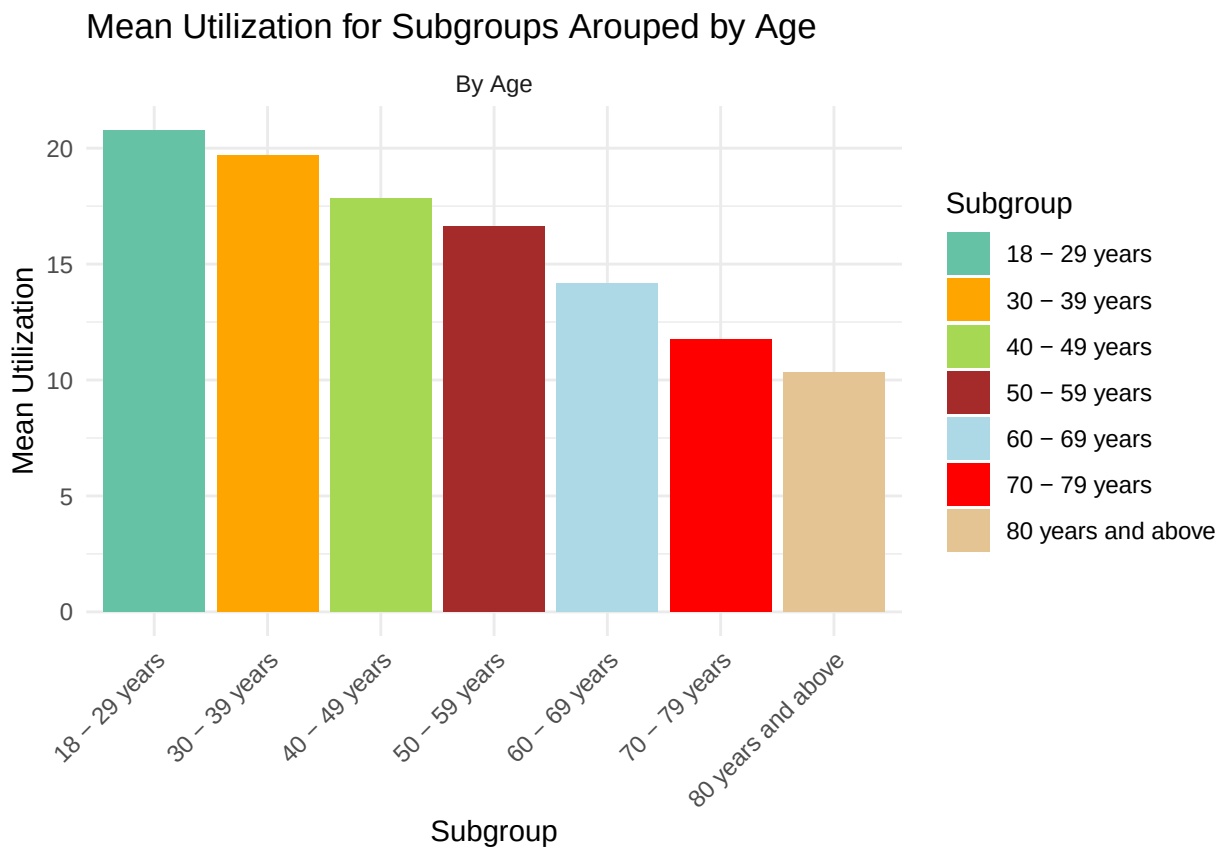


Filter for specific subgroups by age and visualize it.

```
filtered_data <- top_subgroups %>%
  filter(Subgroup %in% c("18 - 29 years", "30 - 39 years", "40 - 49 years", "50 - 59 years", "60 - 69 y

ggplot(filtered_data, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subgrou
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
```

```
  theme_minimal() +
  labs(title = "Mean Utilization for Subgroups Arouped by Age",
       x = "Subgroup",
       y = "Mean Utilization") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = subgroup_colors)  # Keep custom colors
```



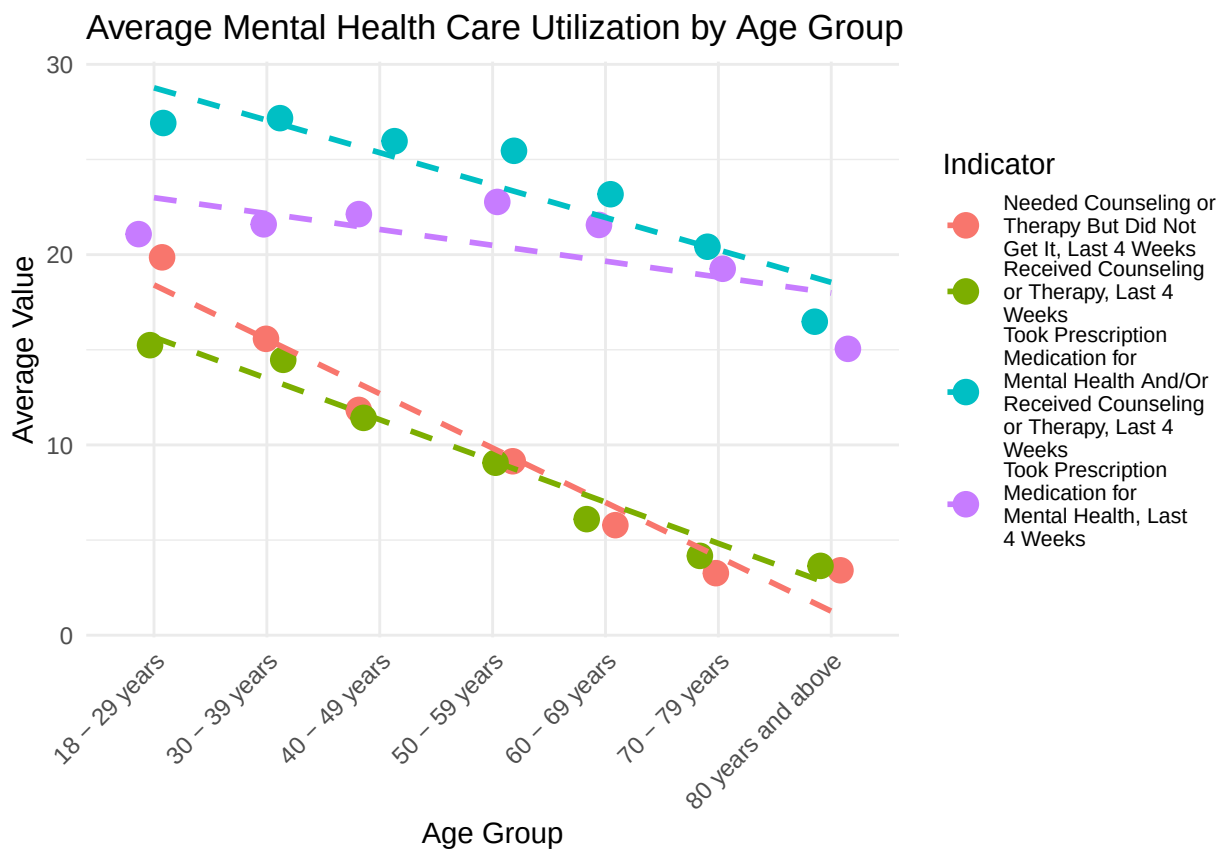
Age group analysis and the different indicators and the visualization of the age group and indicators and their average utlizations.

```
# Compute the average values for each age group and indicator
age_data <- data %>%
  filter(Group == "By Age") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")

# Scatter plot (no regression since Subgroup is categorical)
ggplot(age_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
  labs(
    title = "Average Mental Health Care Utilization by Age Group",
    x = "Age Group",
    y = "Average Value",
    color = "Indicator"
  ) +  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add tre
  theme_minimal() +
```

```r
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),  # Rotate x labels for readability
    legend.text = element_text(size = 8)  # Adjust legend text size
  ) +
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))  # Wrap legend text
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Linear Regression Model
regression_model_age <- lm(Average_Value ~ Subgroup, data = age_data)
summary(regression_model_age)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = age_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.5098 -6.2807 -0.3091  6.9694  9.0265
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  20.774      3.852   5.393 2.39e-05 ***
## Subgroup30 - 39 years        -1.076      5.448  -0.197   0.8454
```

```
## Subgroup40 - 49 years        -2.931      5.448  -0.538   0.5962
## Subgroup50 - 59 years        -4.162      5.448  -0.764   0.4534
## Subgroup60 - 69 years        -6.619      5.448  -1.215   0.2379
## Subgroup70 - 79 years        -9.001      5.448  -1.652   0.1134
## Subgroup80 years and above  -11.135      5.448  -2.044   0.0537 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.704 on 21 degrees of freedom
## Multiple R-squared:  0.2459, Adjusted R-squared:  0.0305
## F-statistic: 1.142 on 6 and 21 DF,  p-value: 0.3734
```

```r
# Correlation analysis
cor_test <- cor.test(age_data$Average_Value, as.numeric(as.factor(age_data$Subgroup)), method = "spearma
```

```
## Warning in cor.test.default(age_data$Average_Value,
## as.numeric(as.factor(age_data$Subgroup)), : Cannot compute exact p-value with
## ties
```

```r
print(cor_test)
```

```
##
##  Spearman's rank correlation rho
##
## data:  age_data$Average_Value and as.numeric(as.factor(age_data$Subgroup))
## S = 5382.6, p-value = 0.01101
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## -0.473081
```

```r
# Checking normality assumption
shapiro_test3 <- shapiro.test(age_data$Average_Value)  # If p-value < 0.05, data is NOT normal
print(shapiro_test3)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  age_data$Average_Value
## W = 0.92788, p-value = 0.05452
```

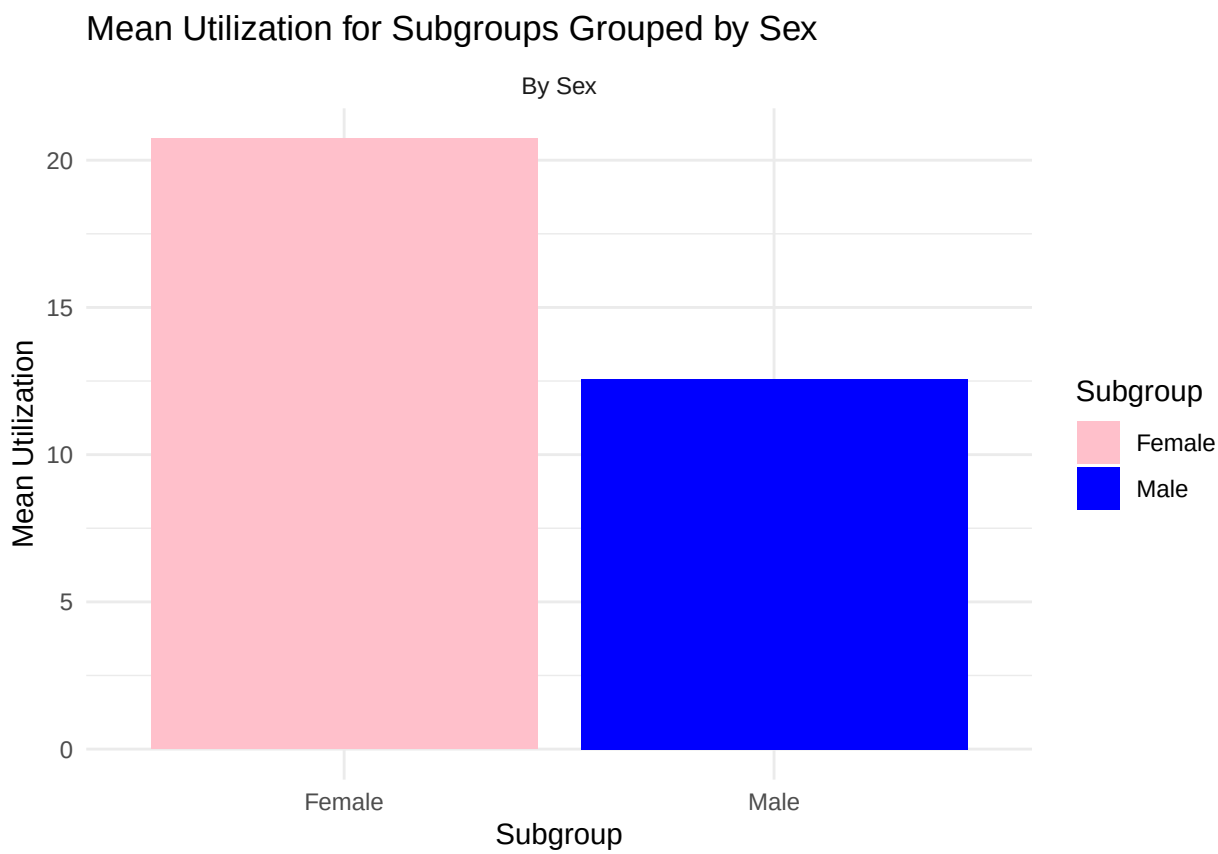
```r
# Kruskal-Wallis Test (since `Subgroup` is categorical)
kruskal_test3 <- kruskal.test(Average_Value ~ Subgroup, data = age_data)
print(kruskal_test3)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Average_Value by Subgroup
## Kruskal-Wallis chi-squared = 6.399, df = 6, p-value = 0.38
```

Filter for specific subgroups by sex and visualize it.

```
filtered_data2 <- top_subgroups %>%
  filter(Subgroup %in% c("Male", "Female"))

ggplot(filtered_data2, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subgr
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
  theme_minimal() +
  labs(title = "Mean Utilization for Subgroups Grouped by Sex",
       x = "Subgroup",
       y = "Mean Utilization") +
  theme(axis.text.x = element_text(angle = 0, hjust = 0.5)) +
  scale_fill_manual(values = subgroup_colors)  # Keep custom colors
```



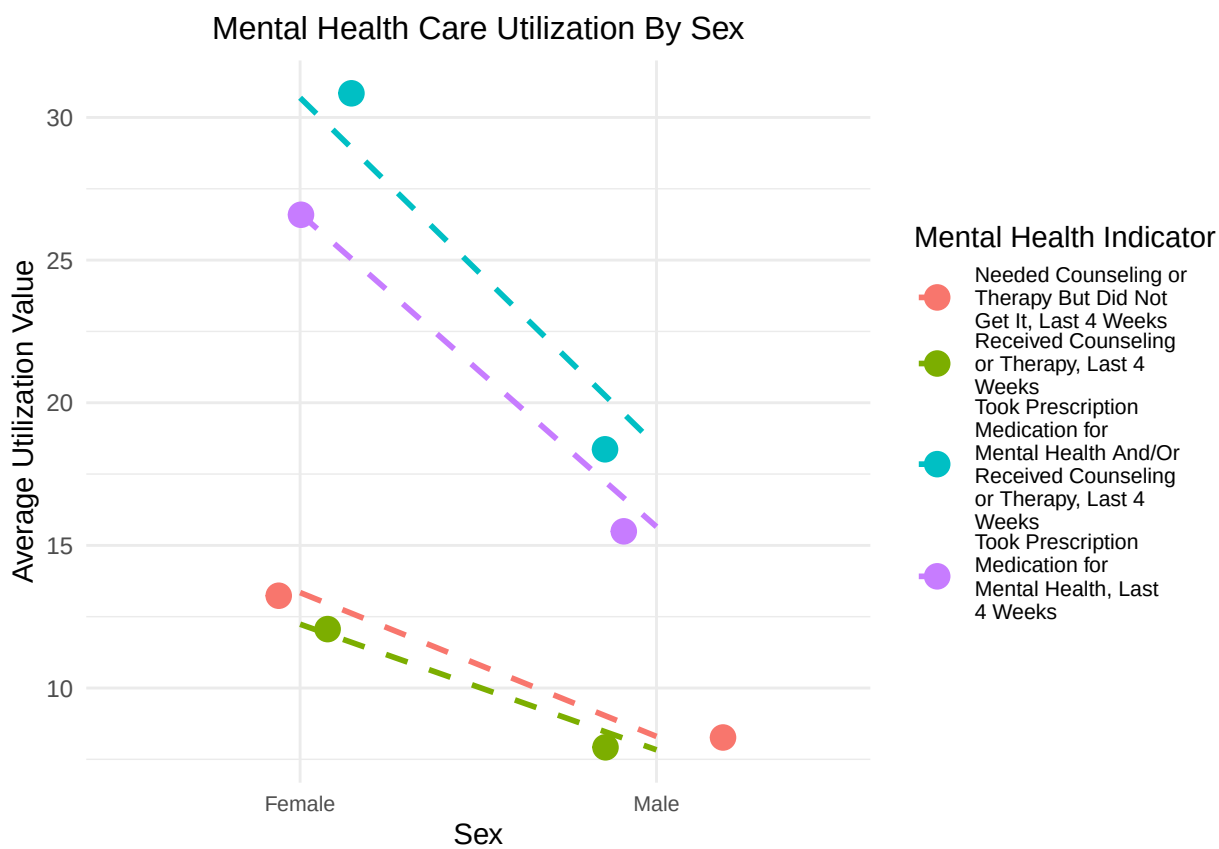
Sex analysis and the different indicators average utilization and its visualization.

```
sex_data <- data %>%
  filter(Group == "By Sex") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")

# Scatter plot
ggplot(sex_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
  theme_minimal() +
```

```r
  labs(
    title = "Mental Health Care Utilization By Sex",
    x = "Sex",
    y = "Average Utilization Value",
    color = "Mental Health Indicator"
  ) +  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add tre
  theme(plot.title = element_text(size = 12, hjust = 0.5),
        axis.text.x = element_text(angle = 0, hjust = .5, size = 8),
        legend.text = element_text(size = 8)) +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 15)) +
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Linear Regression Model
sex_data$Subgroup <- as.factor(sex_data$Subgroup)
sex_model <- lm(Average_Value ~ Subgroup, data = sex_data)
summary(sex_model)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = sex_data)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -8.4886 -5.3983 -0.5924  5.9074  9.9659
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    20.725      3.796   5.459  0.00157 **
## SubgroupMale   -8.154      5.369  -1.519  0.17965
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.593 on 6 degrees of freedom
## Multiple R-squared:  0.2777, Adjusted R-squared:  0.1573
## F-statistic: 2.306 on 1 and 6 DF,  p-value: 0.1796
```

```r
# Correlation analysis
cor_test2 <- cor.test(sex_data$Average_Value, as.numeric(as.factor(sex_data$Subgroup)), method = "spearm
```

```
## Warning in cor.test.default(sex_data$Average_Value,
## as.numeric(as.factor(sex_data$Subgroup)), : Cannot compute exact p-value with
## ties
```

```r
print(cor_test2)
```

```
##
##  Spearman's rank correlation rho
##
## data:  sex_data$Average_Value and as.numeric(as.factor(sex_data$Subgroup))
## S = 120.66, p-value = 0.2797
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.4364358
```

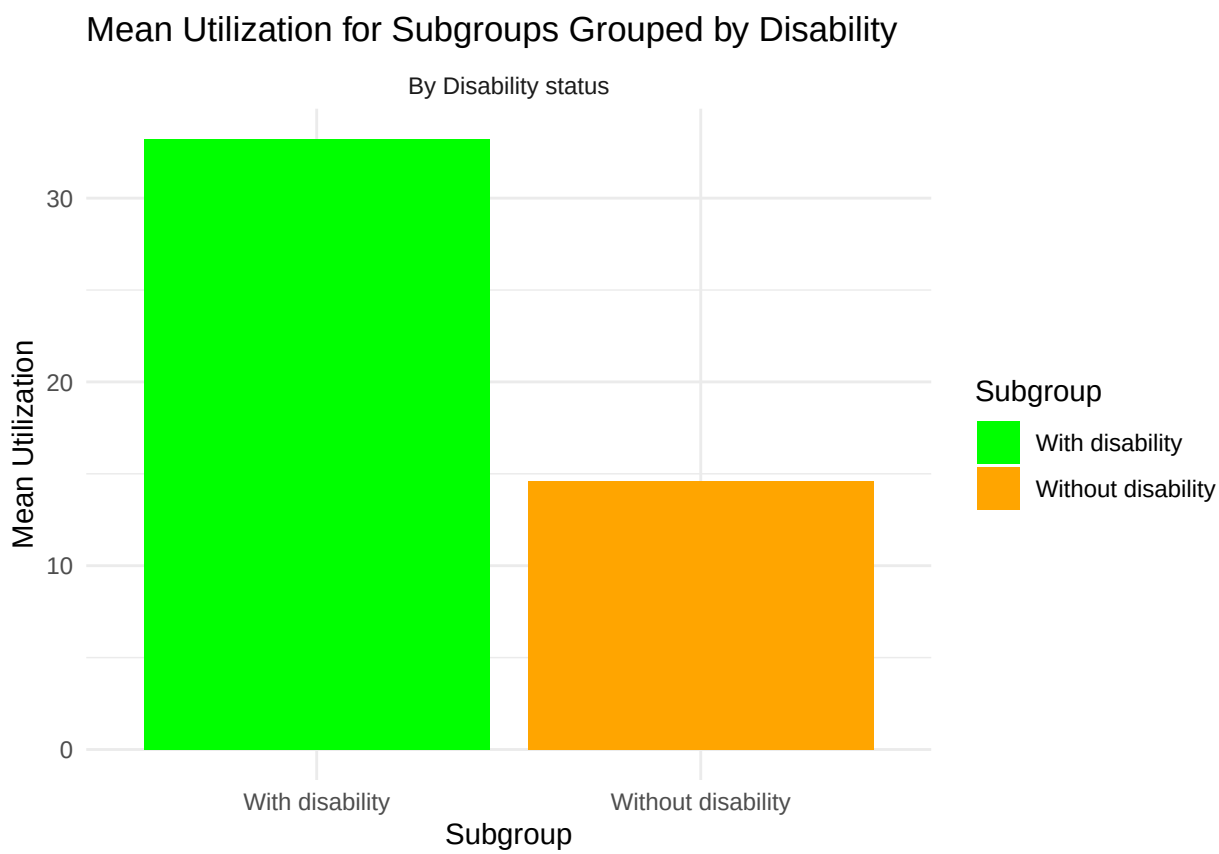
```r
#hypothesis testing
if (length(unique(sex_data$Subgroup)) == 2) {
  t_test_result <- t.test(Average_Value ~ Subgroup, data = sex_data)
  print(t_test_result)
} else {
  anova_result <- aov(Average_Value ~ Subgroup, data = sex_data)
  print(summary(anova_result))
}
```

```
##
##  Welch Two Sample t-test
##
## data:  Average_Value by Subgroup
## t = 1.5187, df = 4.7727, p-value = 0.192
## alternative hypothesis: true difference in means between group Female and group Male is not equal to
## 95 percent confidence interval:
##  -5.847916 22.155492
## sample estimates:
## mean in group Female   mean in group Male
##             20.72500             12.57121
```

Filter for specific subgroups by disability and visualize only the filtered subgroups.

```
filtered_data3 <- top_subgroups %>%
  filter(Subgroup %in% c("With disability", "Without disability"))

ggplot(filtered_data3, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subgr
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
  theme_minimal() +
  labs(title = "Mean Utilization for Subgroups Grouped by Disability",
       x = "Subgroup",
       y = "Mean Utilization") +
  theme(axis.text.x = element_text(angle = 0, hjust = 0.5)) +
  scale_fill_manual(values = subgroup_colors)  # Keep custom colors
```



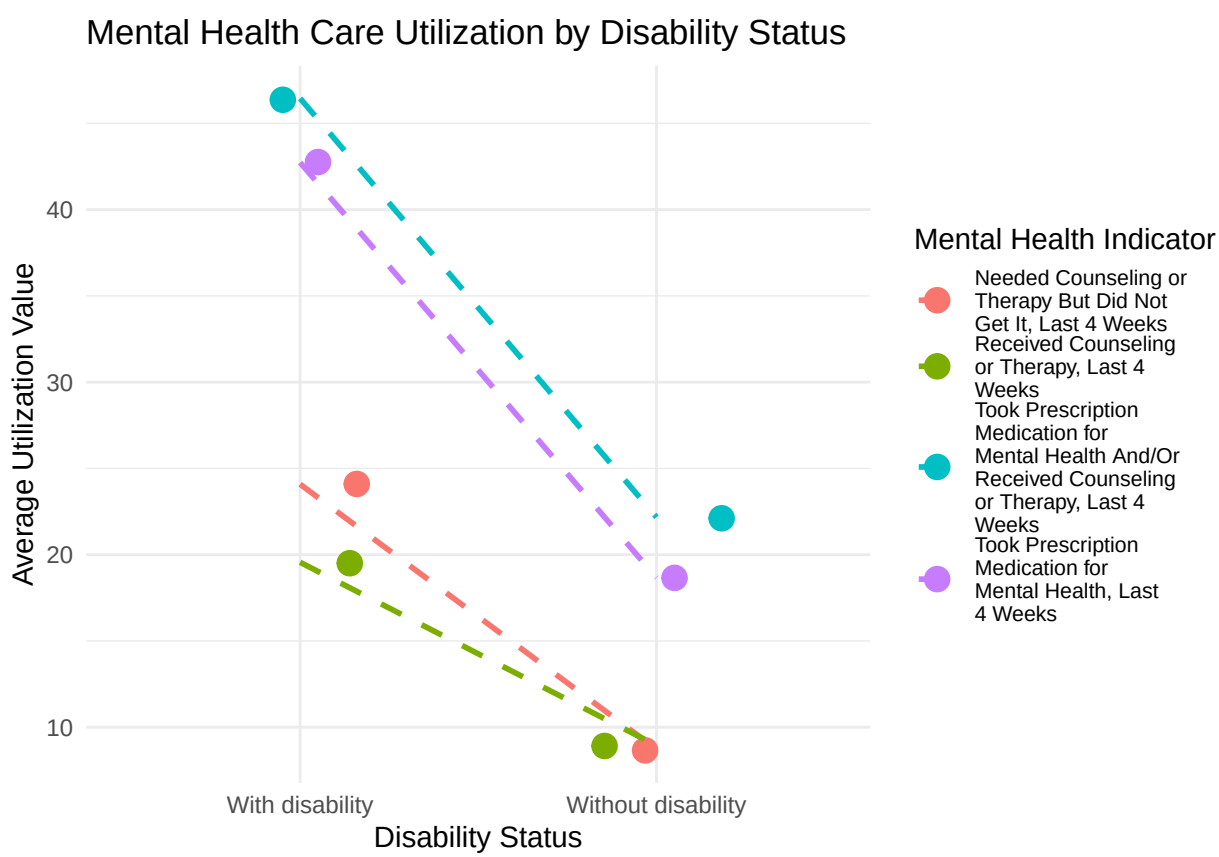
Disability analysis and the different indicators average utilization and their visualization.

```
# Filter and aggregate the data
disability_data <- data %>%
  filter(Group == "By Disability status") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")

# Scatter plot (no regression since `Subgroup` is categorical)
ggplot(disability_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
```

```r
  theme_minimal() +
  labs(
    title = "Mental Health Care Utilization by Disability Status",
    x = "Disability Status",
    y = "Average Utilization Value",
    color = "Mental Health Indicator"
  ) +  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add tre
  theme(
    axis.text.x = element_text(angle = 0, hjust = .5),  # Rotate x labels
    legend.text = element_text(size = 8)  # Adjust legend text size
  ) +
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))  # Wrap legend text
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Linear Regression Model
disability_data$Subgroup <- as.factor(disability_data$Subgroup)
disability_model <- lm(Average_Value ~ Subgroup, data = disability_data)
summary(disability_model)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = disability_data)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.6431  -6.7052  -0.8222   8.0392  13.2403
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  33.204      5.306   6.258 0.000773 ***
## SubgroupWithout disability  -18.576      7.504  -2.476 0.048089 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.61 on 6 degrees of freedom
## Multiple R-squared:  0.5053, Adjusted R-squared:  0.4229
## F-statistic: 6.129 on 1 and 6 DF,  p-value: 0.04809
```

```r
# Correlation analysis
cor_test3 <- cor.test(disability_data$Average_Value, as.numeric(as.factor(disability_data$Subgroup)), me
```

```
## Warning in cor.test.default(disability_data$Average_Value,
## as.numeric(as.factor(disability_data$Subgroup)), : Cannot compute exact p-value
## with ties
```

```r
print(cor_test3)
```

```
##
##  Spearman's rank correlation rho
##
## data:  disability_data$Average_Value and as.numeric(as.factor(disability_data$Subgroup))
## S = 148.16, p-value = 0.0274
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.7637626
```

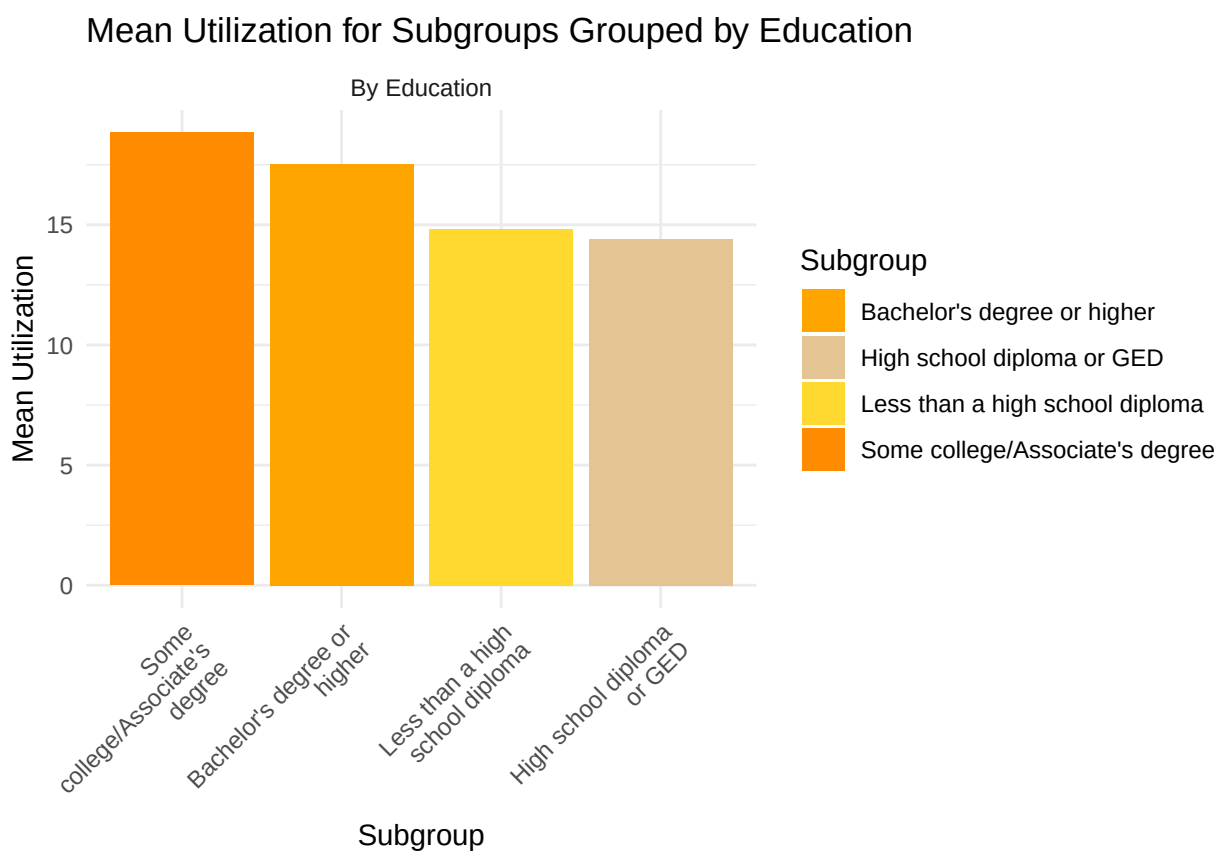
```r
# T-test if only 2 groups exist, otherwise ANOVA
if (length(unique(disability_data$Subgroup)) == 2) {
  t_test_result1 <- t.test(Average_Value ~ Subgroup, data = disability_data)
  print(t_test_result1)
} else {
  anova_result1 <- aov(Average_Value ~ Subgroup, data = disability_data)
  print(summary(anova_result1))
}
```

```
##
##  Welch Two Sample t-test
##
## data:  Average_Value by Subgroup
## t = 2.4756, df = 4.4718, p-value = 0.06196
## alternative hypothesis: true difference in means between group With disability and group Without disa
## 95 percent confidence interval:
##  -1.418351 38.571129
## sample estimates:
##    mean in group With disability mean in group Without disability
##                         33.20417                         14.62778
```

Filter for specific subgroups by education and visualize it.

```
filtered_data4 <- top_subgroups %>%
  filter(Subgroup %in% c("Some college/Associate's degree", "Bachelor's degree or higher", "Less than a

ggplot(filtered_data4, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subgr
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
  theme_minimal() +
  labs(title = "Mean Utilization for Subgroups Grouped by Education",
       x = "Subgroup",
       y = "Mean Utilization") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 20)) +  # Wrap long labels
  scale_fill_manual(values = subgroup_colors)  # Keep custom colors
```



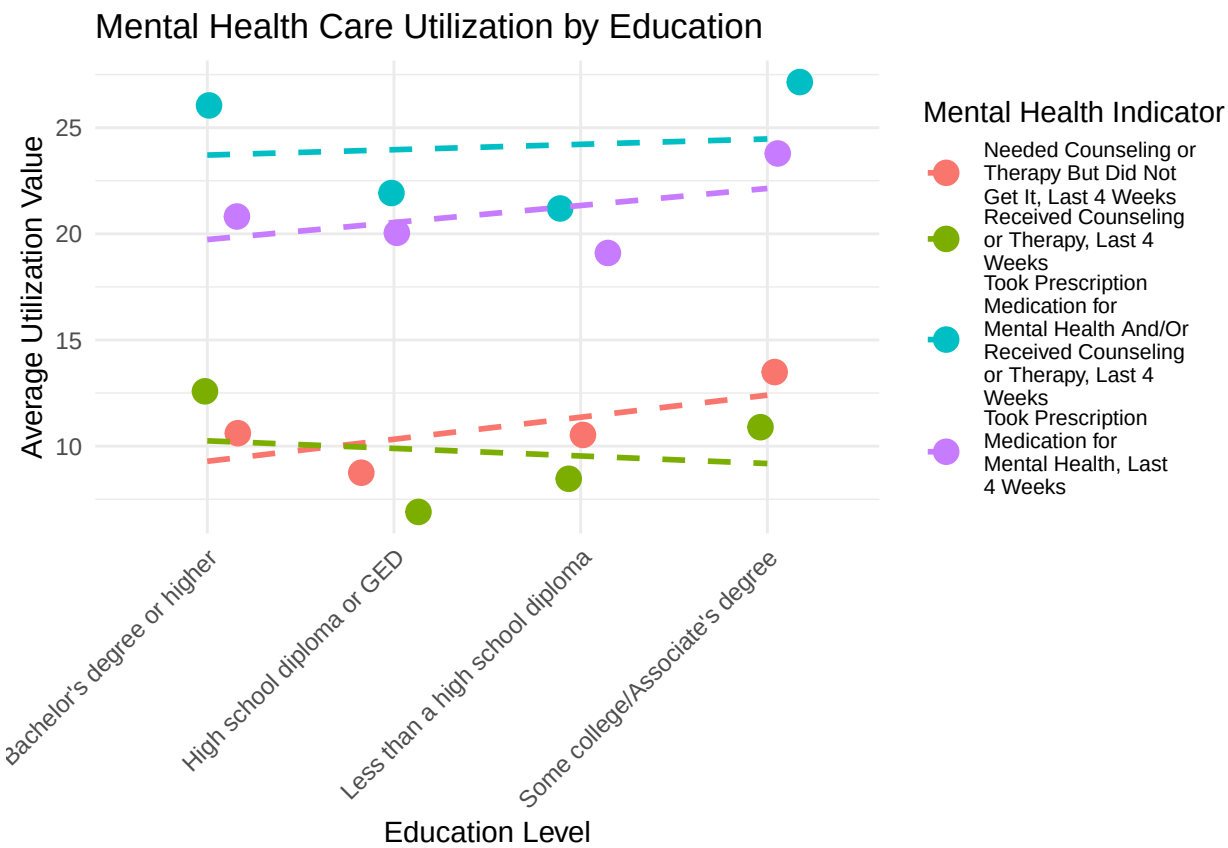
Education analysis and the different indicators average utilization and their visualization.

```
education_data <- data %>%
  filter(Group == "By Education") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")

# Scatter plot
ggplot(education_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
```

```
  theme_minimal() +
  labs(
    title = "Mental Health Care Utilization by Education",
    x = "Education Level",
    y = "Average Utilization Value",
    color = "Mental Health Indicator"
  )  +  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add tr
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.text = element_text(size = 8)) +
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Correlation analysis
cor_test4 <- cor.test(education_data$Average_Value, as.numeric(as.factor(education_data$Subgroup)), met
```

```
## Warning in cor.test.default(education_data$Average_Value,
## as.numeric(as.factor(education_data$Subgroup)), : Cannot compute exact p-value
## with ties
```

```
print(cor_test4)
```

```
##
```

```
##  Spearman's rank correlation rho
##
## data:  education_data$Average_Value and as.numeric(as.factor(education_data$Subgroup))
## S = 605.78, p-value = 0.6874
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.109141
```

```r
# Regression and normality tests
education_data$Subgroup <- as.factor(education_data$Subgroup)
education_model <- lm(Average_Value ~ Subgroup, data = education_data)
summary(education_model)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = education_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -7.926 -5.841 -0.497  5.815  8.549
##
## Coefficients:
##                                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                                17.520      3.639   4.814 0.000423
## SubgroupHigh school diploma or GED         -3.113      5.147  -0.605 0.556560
## SubgroupLess than a high school diploma    -2.695      5.147  -0.524 0.610010
## SubgroupSome college/Associate's degree     1.309      5.147   0.254 0.803529
##
## (Intercept)                              ***
## SubgroupHigh school diploma or GED
## SubgroupLess than a high school diploma
## SubgroupSome college/Associate's degree
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.279 on 12 degrees of freedom
## Multiple R-squared:  0.07887,    Adjusted R-squared:  -0.1514
## F-statistic: 0.3425 on 3 and 12 DF,  p-value: 0.7951
```

```r
if (nrow(education_data) < 5000) {
  shapiro.test(education_data$Average_Value)
} else {
  ks.test(education_data$Average_Value, "pnorm", mean(education_data$Average_Value), sd(education_data$A
}
```

```
##
##  Shapiro-Wilk normality test
##
## data:  education_data$Average_Value
## W = 0.91362, p-value = 0.1331
```

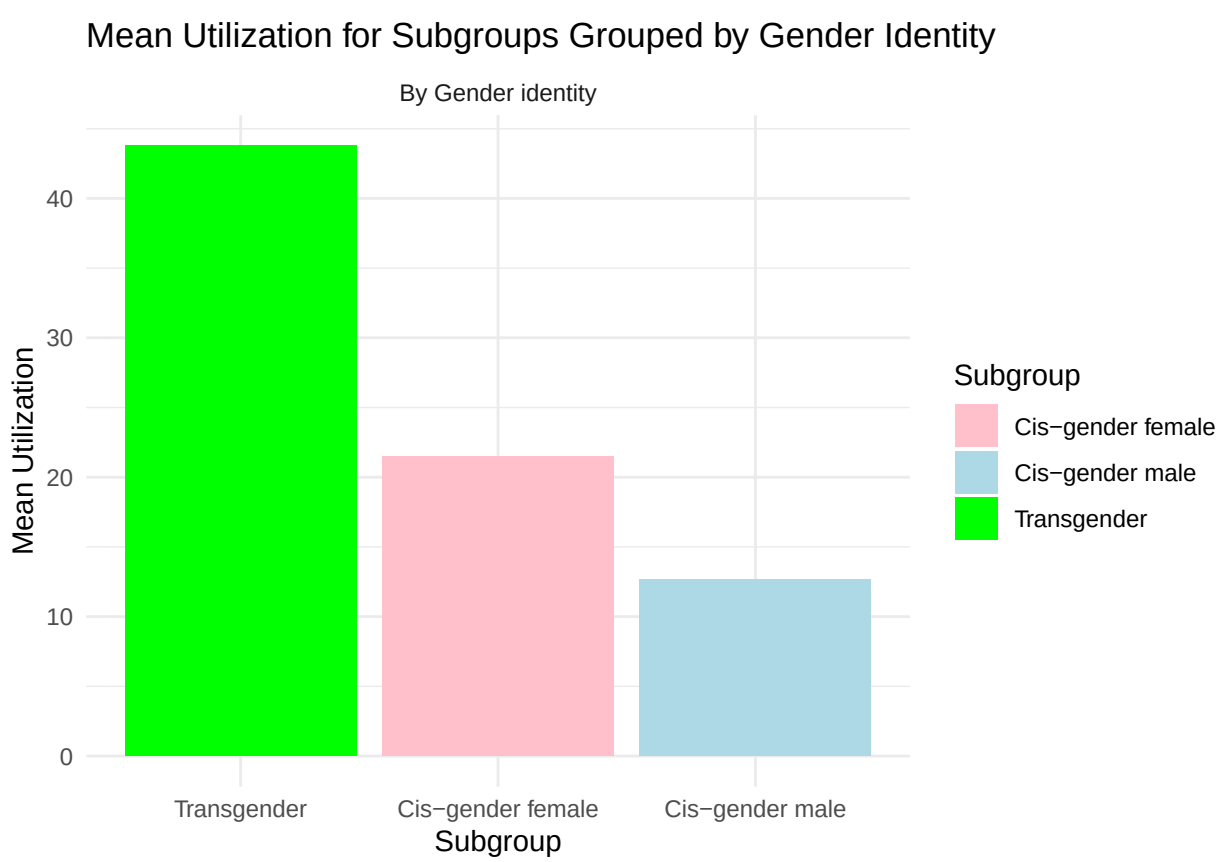
```r
kruskal_test4 <- kruskal.test(Average_Value ~ Subgroup, data = education_data)
print(kruskal_test4)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Average_Value by Subgroup
## Kruskal-Wallis chi-squared = 2.3603, df = 3, p-value = 0.5011
```

Filter for specific subgroups by gender identity and visualize it.

```r
filtered_data5 <- top_subgroups %>%
  filter(Subgroup %in% c("Cis-gender female", "Cis-gender male", "Transgender"))

ggplot(filtered_data5, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subgro
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
  theme_minimal() +
  labs(title = "Mean Utilization for Subgroups Grouped by Gender Identity",
       x = "Subgroup",
       y = "Mean Utilization") +
  theme(axis.text.x = element_text(angle = 0, hjust = 0.5)) +
  scale_fill_manual(values = subgroup_colors)  # Keep custom colors
```



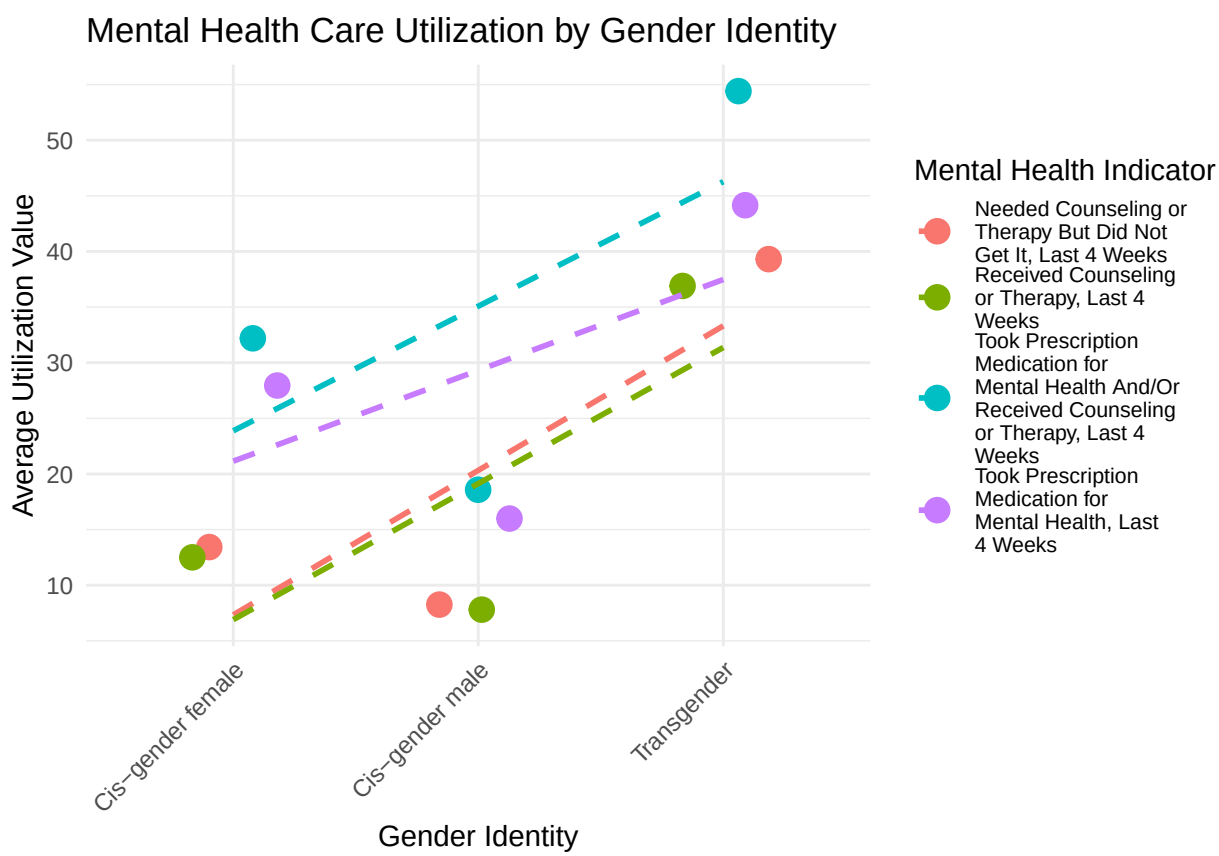
Gender identity analysis and the different indicators average utilization and their visualization.

```r
gender_data <- data %>%
  filter(Group == "By Gender identity") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")

# Scatter plot
ggplot(gender_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
  theme_minimal() +
  labs(
    title = "Mental Health Care Utilization by Gender Identity",
    x = "Gender Identity",
    y = "Average Utilization Value",
    color = "Mental Health Indicator"
  ) +  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add tre

  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.text = element_text(size = 8)) +
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# regression model
gender_data$Subgroup <- as.factor(gender_data$Subgroup)
```

```r
gender_model <- lm(Average_Value ~ Subgroup, data = gender_data)
summary(gender_model)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = gender_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.902 -5.317 -1.977  6.157 10.685
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                21.477      3.968   5.412 0.000426 ***
## SubgroupCis-gender male    -8.840      5.612  -1.575 0.149686
## SubgroupTransgender        22.296      5.612   3.973 0.003241 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.937 on 9 degrees of freedom
## Multiple R-squared:  0.7842, Adjusted R-squared:  0.7362
## F-statistic: 16.35 on 2 and 9 DF,  p-value: 0.001008
```

```r
# Correlation analysis
cor_test5 <- cor.test(gender_data$Average_Value, as.numeric(as.factor(gender_data$Subgroup)), method =
```

```
## Warning in cor.test.default(gender_data$Average_Value,
## as.numeric(as.factor(gender_data$Subgroup)), : Cannot compute exact p-value
## with ties
```

```r
print(cor_test5)
```

```
##
##  Spearman's rank correlation rho
##
## data:  gender_data$Average_Value and as.numeric(as.factor(gender_data$Subgroup))
## S = 116.88, p-value = 0.04287
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.5913124
```

```r
# hypothesis testing
if (nrow(gender_data) < 5000) {
  shapiro.test(gender_data$Average_Value)
} else {
  ks.test(gender_data$Average_Value, "pnorm", mean(gender_data$Average_Value), sd(gender_data$Average_Va
}
```

```
##
##  Shapiro-Wilk normality test
```

```
##
## data:  gender_data$Average_Value
## W = 0.9272, p-value = 0.3513
```

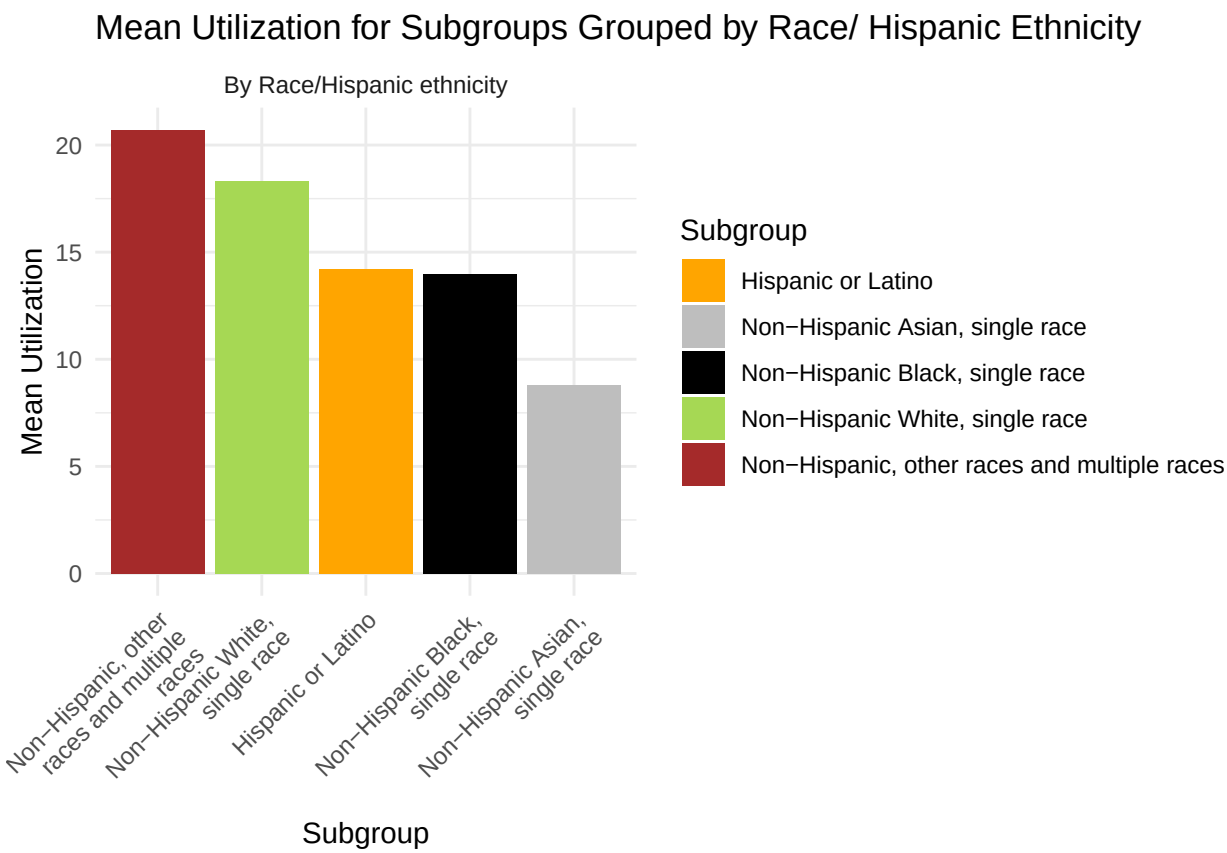
```r
kruskal_test5 <- kruskal.test(Average_Value ~ Subgroup, data = gender_data)
print(kruskal_test5)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Average_Value by Subgroup
## Kruskal-Wallis chi-squared = 8, df = 2, p-value = 0.01832
```

Filter for specific subgroups by race/ ethinicity and visualize it.

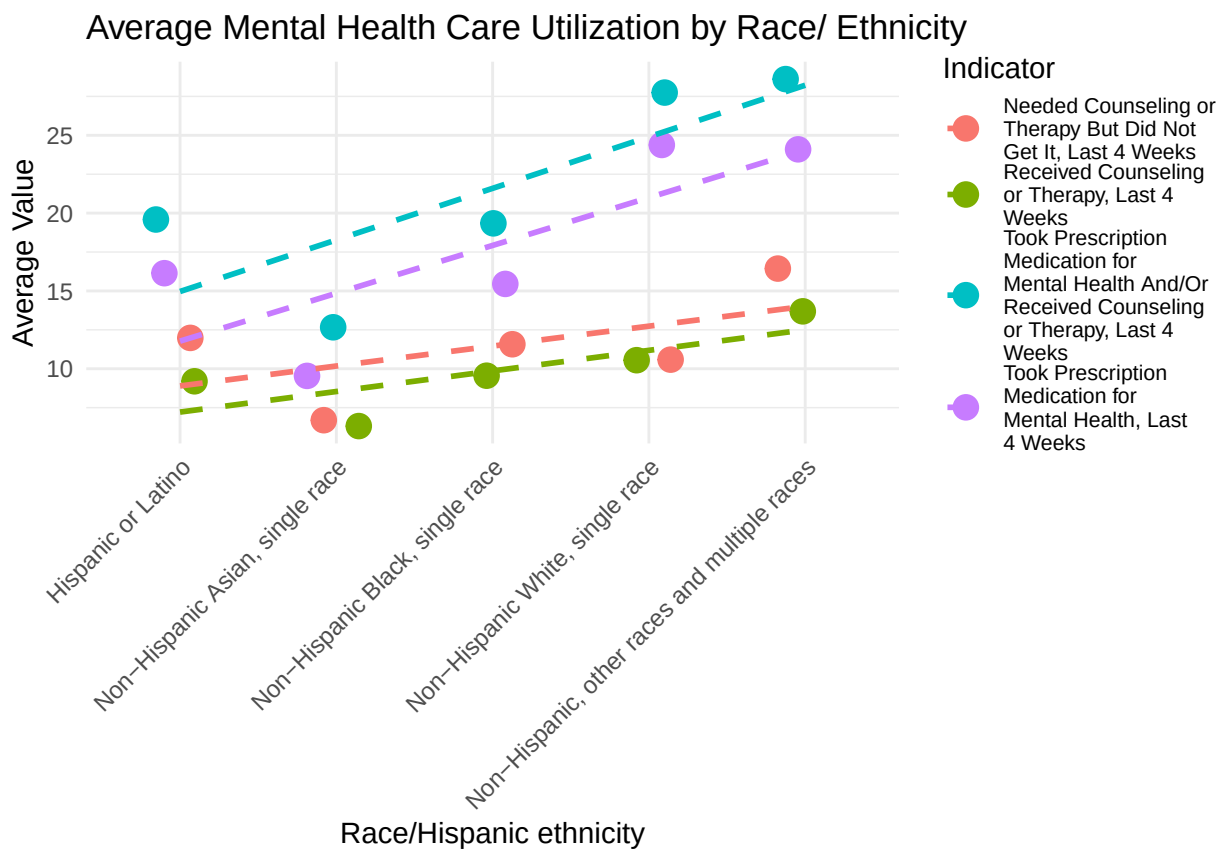
```r
filtered_data6 <- top_subgroups %>%
  filter(Subgroup %in% c("Non-Hispanic, other races and multiple races", "Non-Hispanic White, single rac

ggplot(filtered_data6, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subgro
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
  theme_minimal() +
  labs(title = "Mean Utilization for Subgroups Grouped by Race/ Hispanic Ethnicity",
       x = "Subgroup",
       y = "Mean Utilization") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 20)) +  # Wrap long labels
  scale_fill_manual(values = subgroup_colors)  # Keep custom colors
```




```r
# Compute the average values for each age group and indicator
# Filter and aggregate the data
race_data <- data %>%
  filter(Group == "By Race/Hispanic ethnicity") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")
# Scatter plot (no regression since Subgroup is categorical)
ggplot(race_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
  labs(
    title = "Average Mental Health Care Utilization by Race/ Ethnicity",
    x = "Race/Hispanic ethnicity",
    y = "Average Value",
    color = "Indicator"
  ) +  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add tre
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),  # Rotate x labels for readability
    legend.text = element_text(size = 8)  # Adjust legend text size
  ) +
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))  # Wrap legend text
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Linear Regression Model
regression_model_race <- lm(Average_Value ~ Subgroup, data = race_data)
summary(regression_model_race)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = race_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.7712 -4.3078 -0.6886  4.2388  9.4379
##
## Coefficients:
##                                                    Estimate Std. Error
## (Intercept)                                         14.2288     2.9790
## SubgroupNon-Hispanic Asian, single race             -5.4265     4.2129
## SubgroupNon-Hispanic Black, single race             -0.2447     4.2129
## SubgroupNon-Hispanic White, single race              4.0848     4.2129
## SubgroupNon-Hispanic, other races and multiple races   6.4818     4.2129
##                                                    t value Pr(>|t|)
## (Intercept)                                          4.776 0.000245 ***
## SubgroupNon-Hispanic Asian, single race             -1.288 0.217240
## SubgroupNon-Hispanic Black, single race             -0.058 0.954450
## SubgroupNon-Hispanic White, single race              0.970 0.347622
## SubgroupNon-Hispanic, other races and multiple races   1.539 0.144738
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.958 on 15 degrees of freedom
## Multiple R-squared:  0.3852, Adjusted R-squared:  0.2213
## F-statistic:  2.35 on 4 and 15 DF,  p-value: 0.1012
```

```r
# Correlation analysis
cor_test6 <- cor.test(race_data$Average_Value, as.numeric(as.factor(race_data$Subgroup)), method = "spea
```

```
## Warning in cor.test.default(race_data$Average_Value,
## as.numeric(as.factor(race_data$Subgroup)), : Cannot compute exact p-value with
## ties
```

```r
print(cor_test6)
```

```
##
##  Spearman's rank correlation rho
##
## data:  race_data$Average_Value and as.numeric(as.factor(race_data$Subgroup))
## S = 718.39, p-value = 0.04135
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.4598545
```

```r
# Checking normality assumption
shapiro_test4 <- shapiro.test(race_data$Average_Value)  # If p-value < 0.05, data is NOT normal
print(shapiro_test4)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  race_data$Average_Value
## W = 0.91771, p-value = 0.08954
```

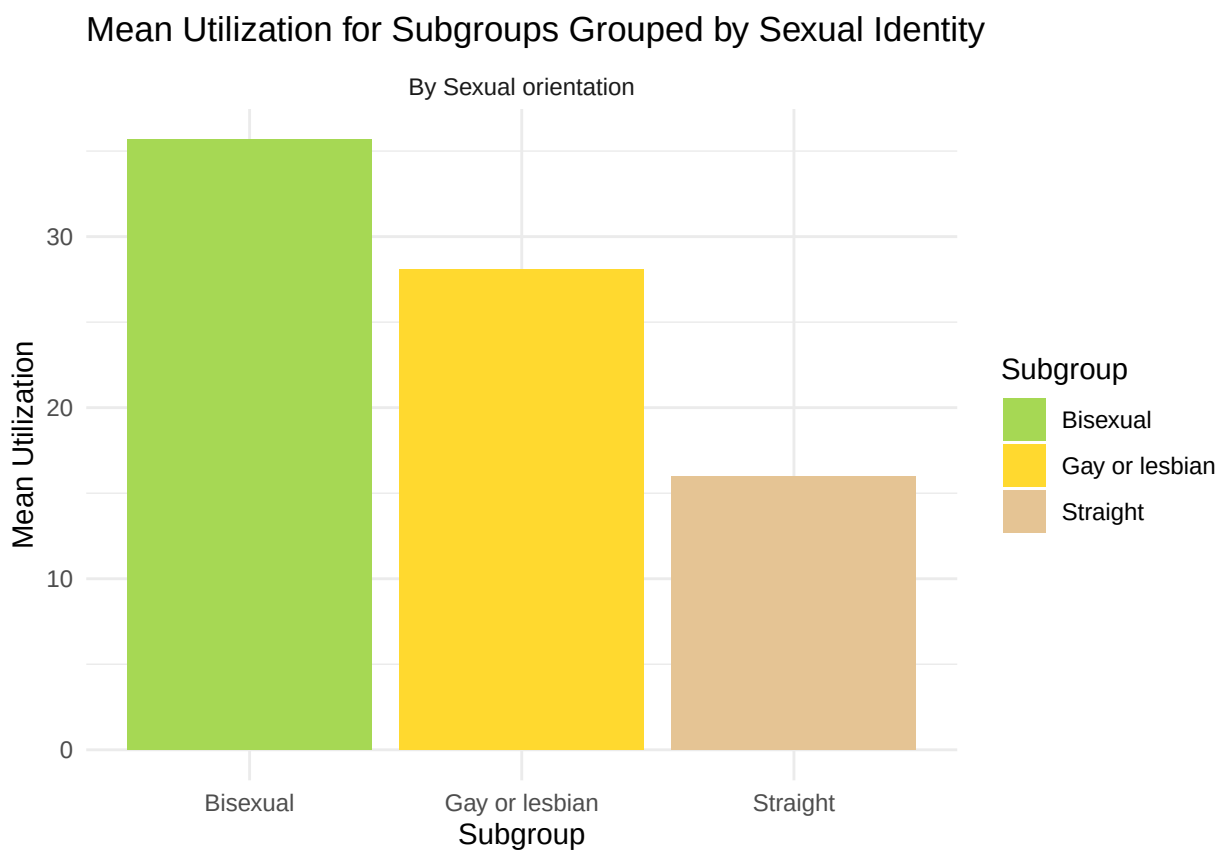
```r
# Kruskal-Wallis Test (since `Subgroup` is categorical)
kruskal_test6 <- kruskal.test(Average_Value ~ Subgroup, data = race_data)
print(kruskal_test6)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Average_Value by Subgroup
## Kruskal-Wallis chi-squared = 7.8143, df = 4, p-value = 0.09862
```

Filter for specific subgroups by sexual identity and visualize it.

```r
filtered_data7 <- top_subgroups %>%
  filter(Subgroup %in% c("Gay or lesbian", "Straight", "Bisexual"))
```

```r
ggplot(filtered_data7, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subgr
  geom_col() +
  facet_wrap(~ Group, scales = "free_y") +
  theme_minimal() +
  labs(title = "Mean Utilization for Subgroups Grouped by Sexual Identity",
       x = "Subgroup",
       y = "Mean Utilization") +
  theme(axis.text.x = element_text(angle = 0, hjust = 0.5)) +
  scale_fill_manual(values = subgroup_colors)  # Keep custom colors
```



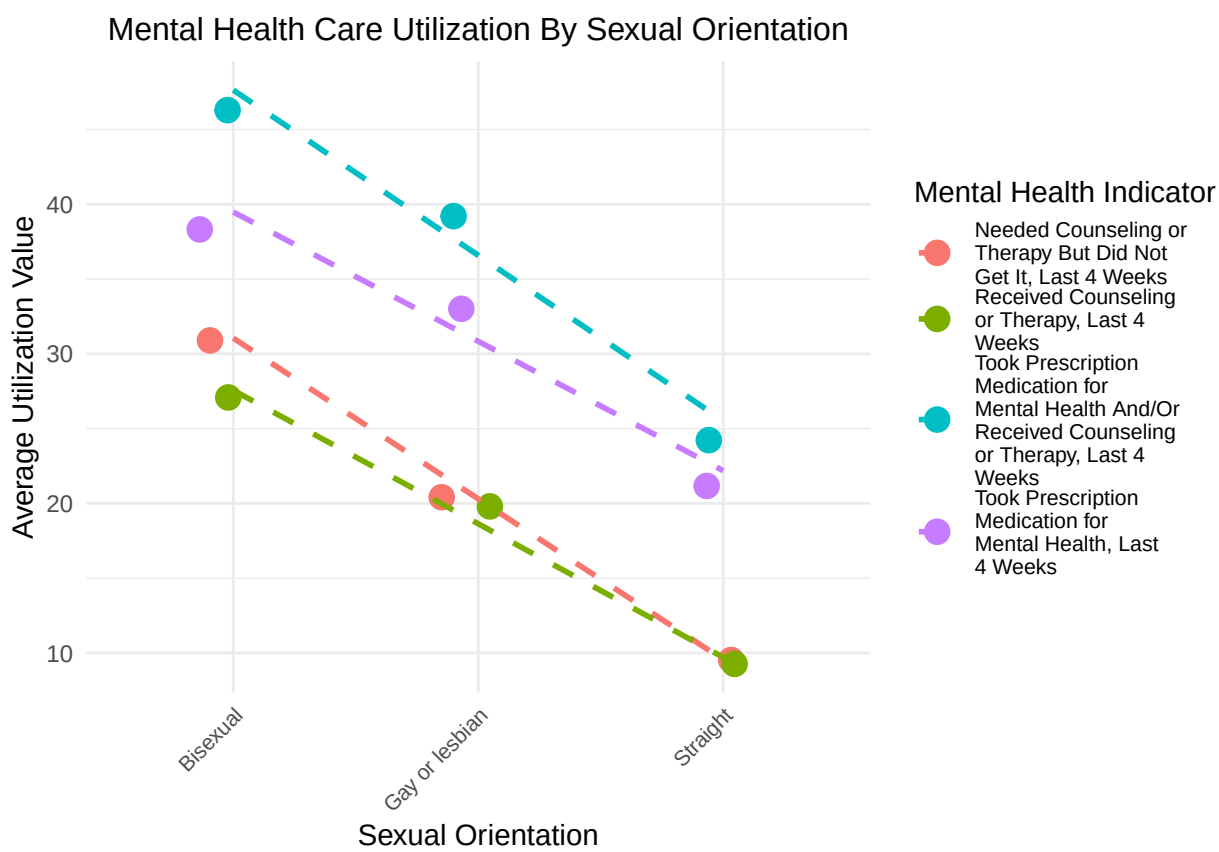
Sexual orientation analysis and the different indicators average utilization and their visualization.

```r
# Grouping data by Sexual orientation and Indicator, then calculating mean value
orientation_data <- data %>%
  filter(Group == "By Sexual orientation") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")

# Scatter plot (no regression since Subgroup is categorical)
ggplot(orientation_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add trendlin
  theme_minimal() +
  labs(
    title = "Mental Health Care Utilization By Sexual Orientation",
    x = "Sexual Orientation",
```

```r
    y = "Average Utilization Value",
    color = "Mental Health Indicator"
  ) +
  theme(
    plot.title = element_text(size = 12, hjust = 0.5),  # Adjust title size and center it
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),  # Rotate x labels for clarity
    legend.text = element_text(size = 8)  # Adjust legend text size
  ) +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 15)) +  # Wrap x-axis labels
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))  # Wrap legend text
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Convert Subgroup to Factor
orientation_data$Subgroup <- as.factor(orientation_data$Subgroup)

# Linear Regression Model
regression_model_orientation <- lm(Average_Value ~ Subgroup, data = orientation_data)
summary(regression_model_orientation)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = orientation_data)
##
```

```
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.658 -7.033 -1.017  5.862 11.017
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)              35.675      4.336   8.228 1.77e-05 ***
## SubgroupGay or lesbian   -7.567      6.132  -1.234   0.2485
## SubgroupStraight        -19.692      6.132  -3.211   0.0106 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.672 on 9 degrees of freedom
## Multiple R-squared:  0.5384, Adjusted R-squared:  0.4358
## F-statistic: 5.248 on 2 and 9 DF,  p-value: 0.03085
```

```r
# Correlation analysis
cor_test7 <- cor.test(orientation_data$Average_Value, as.numeric(as.factor(orientation_data$Subgroup)),
```

```
## Warning in cor.test.default(orientation_data$Average_Value,
## as.numeric(as.factor(orientation_data$Subgroup)), : Cannot compute exact
## p-value with ties
```

```r
print(cor_test7)
```

```
##
##  Spearman's rank correlation rho
##
## data:  orientation_data$Average_Value and as.numeric(as.factor(orientation_data$Subgroup))
## S = 480.48, p-value = 0.01497
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.6800093
```

```r
# Statistical Tests for Normality and Group Differences
shapiro_result <- shapiro.test(orientation_data$Average_Value)
print(shapiro_result)  # Print test results
```

```
##
##  Shapiro-Wilk normality test
##
## data:  orientation_data$Average_Value
## W = 0.96475, p-value = 0.8488
```

```r
if (shapiro_result$p.value < 0.05) {
  # If data is not normally distributed, use Kruskal-Wallis
  kruskal_result <- kruskal.test(Average_Value ~ Subgroup, data = orientation_data)
  print(kruskal_result)
} else {
  # If data is normal, ANOVA could be used (if needed)
```

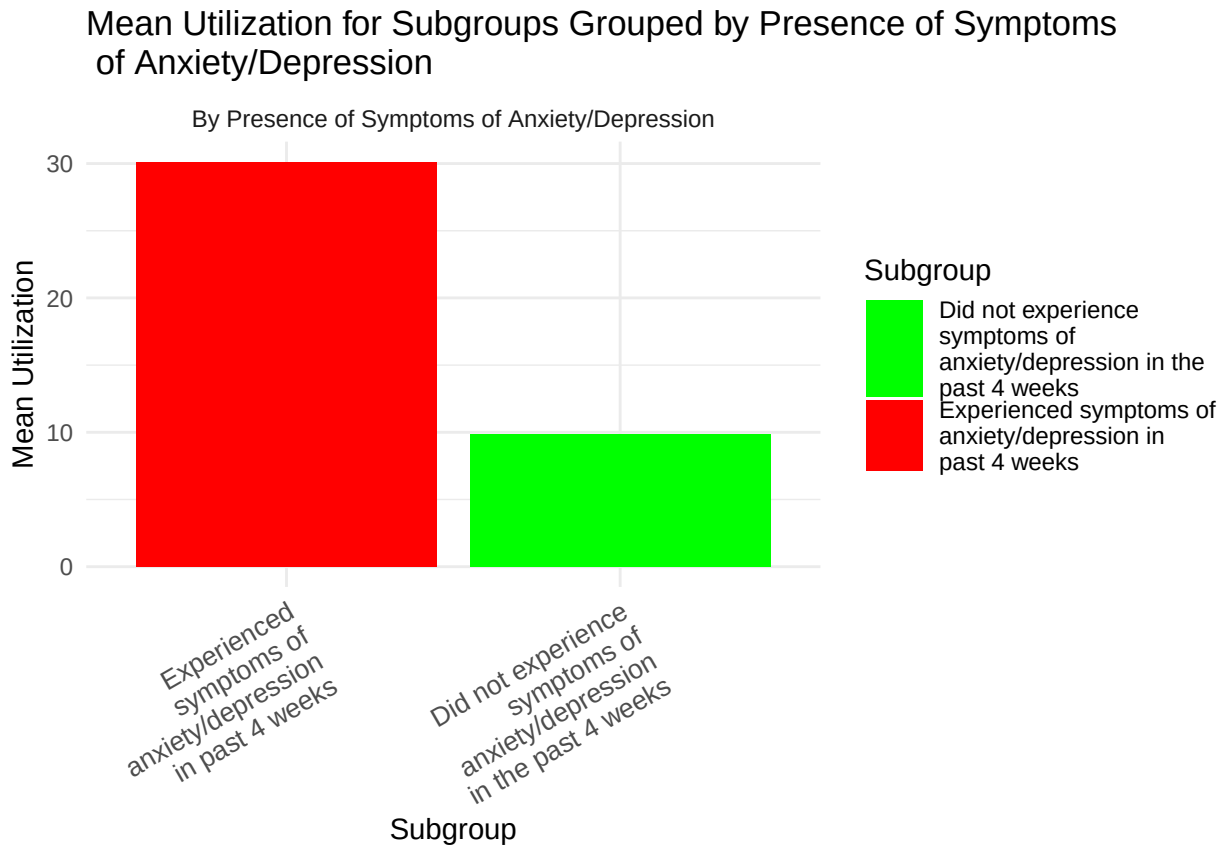
```r
  anova_result2 <- aov(Average_Value ~ Subgroup, data = orientation_data)
  print(summary(anova_result2))
}
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## Subgroup     2  789.4   394.7   5.248 0.0308 *
## Residuals    9  676.8    75.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Filter for specific subgroups by presence of symptoms and visualize it.

```r
filtered_data8 <- top_subgroups %>%
    filter(Subgroup %in% c("Experienced symptoms of anxiety/depression in past 4 weeks",
                           "Did not experience symptoms of anxiety/depression in the past 4 weeks"))

  ggplot(filtered_data8, aes(x = reorder(Subgroup, -mean_utilization), y = mean_utilization, fill = Subg
    geom_col() +
    facet_wrap(~ Group, scales = "free_y") +
    theme_minimal() +
    labs(title = "Mean Utilization for Subgroups Grouped by Presence of Symptoms \n of Anxiety/Depressio
         x = "Subgroup",
         y = "Mean Utilization") +
    theme(
      axis.text.x = element_text(angle = 30, hjust = 1, size = 10),  # Slight rotation for readability
      legend.text = element_text(size = 9),  # Adjust legend text size
      legend.key.size = unit(0.8, "cm"),  # Reduce legend key size
      legend.spacing.y = unit(0.5, "cm")  # Reduce vertical spacing between legend items
    ) +
    scale_x_discrete(labels = function(x) str_wrap(x, width = 20)) +  # Wrap x-axis text
    scale_fill_manual(values = subgroup_colors,
                      labels = function(x) str_wrap(x, width = 25)) +  # Wrap legend text
    guides(fill = guide_legend(ncol = 1))  # Arrange legend items in a single column
```

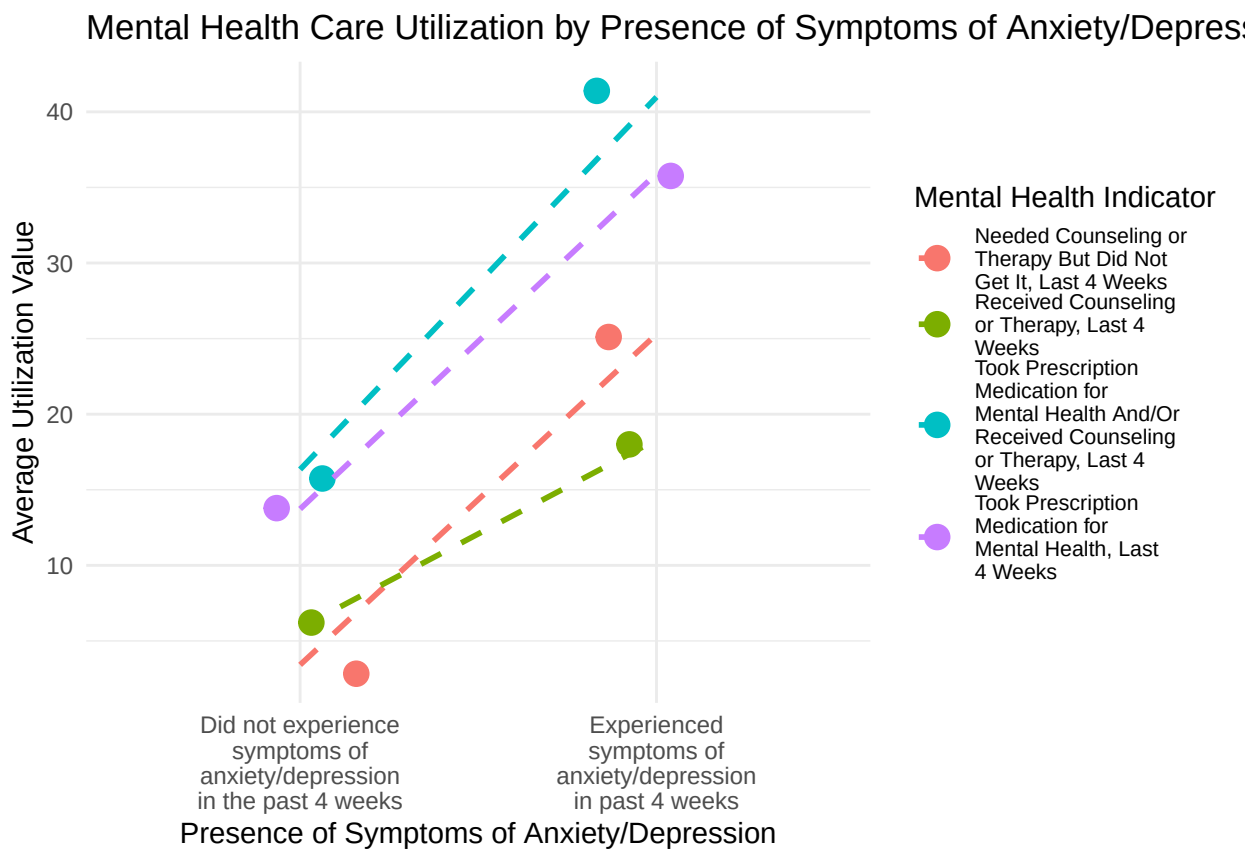


Sympton analysis and the different indicator average utilization and their visualization.

```
# Filter and aggregate the data
symptom_data <- data %>%
  filter(Group == "By Presence of Symptoms of Anxiety/Depression") %>%
  group_by(Subgroup, Indicator) %>%
  summarise(Average_Value = mean(Value, na.rm = TRUE), .groups = "drop")

# Scatter plot (no regression since `Subgroup` is categorical)
ggplot(symptom_data, aes(x = Subgroup, y = Average_Value, color = Indicator)) +
  geom_point(size = 4, position = position_jitter(width = 0.2)) +
  theme_minimal() +
  labs(
    title = "Mental Health Care Utilization by Presence of Symptoms of Anxiety/Depression",
    x = "Presence of Symptoms of Anxiety/Depression",
    y = "Average Utilization Value",
    color = "Mental Health Indicator"
  ) +  geom_smooth(method = "lm", se = FALSE, aes(group = Indicator), linetype = "dashed") +  # Add tre
  theme(
    axis.text.x = element_text(angle = 0, hjust = .5),  # Rotate x labels
    legend.text = element_text(size = 8)  # Adjust legend text size
  ) + scale_x_discrete(labels = function(x) str_wrap(x, width = 20)) +  # Wrap x-axis text
  scale_color_discrete(labels = function(x) stringr::str_wrap(x, width = 20))  # Wrap legend text
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Linear Regression Model
symptom_data$Subgroup <- as.factor(symptom_data$Subgroup)
symptom_model <- lm(Average_Value ~ Subgroup, data = symptom_data)
summary(symptom_model)
```

```
##
## Call:
## lm(formula = Average_Value ~ Subgroup, data = symptom_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.8114  -5.2311  -0.0447   5.9705  10.8553
##
## Coefficients:
##                                                                  Estimate
## (Intercept)                                                         9.836
## SubgroupExperienced symptoms of anxiety/depression in past 4 weeks   20.288
##                                                                  Std. Error
## (Intercept)                                                           4.227
## SubgroupExperienced symptoms of anxiety/depression in past 4 weeks      5.978
##                                                                  t value
## (Intercept)                                                        2.327
## SubgroupExperienced symptoms of anxiety/depression in past 4 weeks   3.394
##                                                                  Pr(>|t|)
## (Intercept)                                                        0.0589 .
## SubgroupExperienced symptoms of anxiety/depression in past 4 weeks   0.0146 *
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.454 on 6 degrees of freedom
## Multiple R-squared:  0.6575, Adjusted R-squared:  0.6004
## F-statistic: 11.52 on 1 and 6 DF,  p-value: 0.0146
```

```r
# Correlation analysis
cor_test8 <- cor.test(symptom_data$Average_Value, as.numeric(as.factor(symptom_data$Subgroup)), method =
```

```
## Warning in cor.test.default(symptom_data$Average_Value,
## as.numeric(as.factor(symptom_data$Subgroup)), : Cannot compute exact p-value
## with ties
```

```r
print(cor_test8)
```

```
##
##  Spearman's rank correlation rho
##
## data:  symptom_data$Average_Value and as.numeric(as.factor(symptom_data$Subgroup))
## S = 10.679, p-value = 0.004659
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.8728716
```

```r
# T-test if only 2 groups exist, otherwise ANOVA
if (length(unique(symptom_data$Subgroup)) == 2) {
  t_test_result2 <- t.test(Average_Value ~ Subgroup, data = symptom_data)
  print(t_test_result2)
} else {
  anova_result3 <- aov(Average_Value ~ Subgroup, data = symptom_data)
  print(summary(anova_result3))
}
```

```
##
##  Welch Two Sample t-test
##
## data:  Average_Value by Subgroup
## t = -3.394, df = 4.93, p-value = 0.01981
## alternative hypothesis: true difference in means between group Did not experience symptoms of anxiet
## 95 percent confidence interval:
##  -35.719746  -4.856012
## sample estimates:
## mean in group Did not experience symptoms of anxiety/depression in the past 4 weeks
##                                                                          9.835606
##            mean in group Experienced symptoms of anxiety/depression in past 4 weeks
##                                                                         30.123485
```

Summarize average utilization throughout the different phases and view the summary.

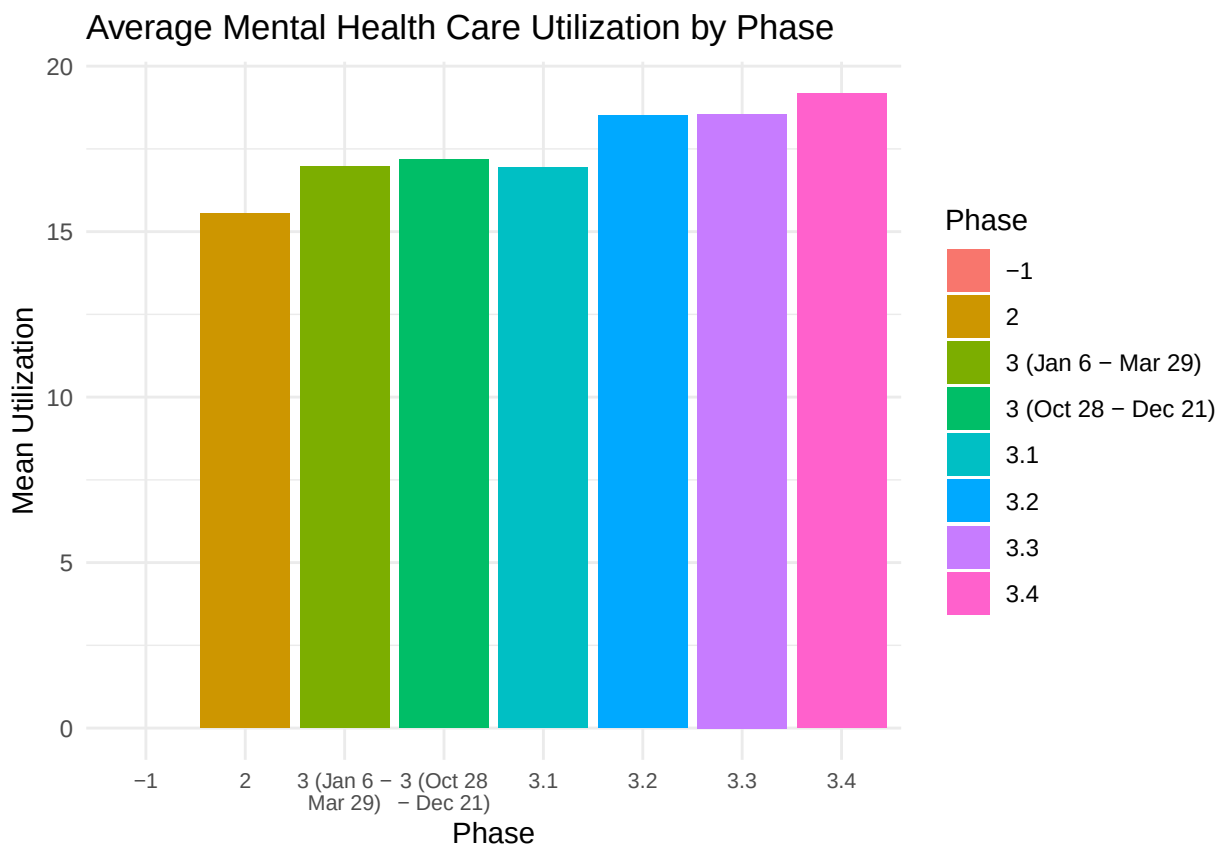
```r
phase_summary <- data %>%
  group_by(Phase) %>%
  summarize(mean_utilization = mean(Value, na.rm = TRUE),
            median_utilization = median(Value, na.rm = TRUE),
            sd_utilization = sd(Value, na.rm = TRUE))
print(phase_summary)
```

```
## # A tibble: 8 x 4
##   Phase                mean_utilization median_utilization sd_utilization
##   <chr>                           <dbl>              <dbl>          <dbl>
## 1 -1                              NaN                 NA            NA
## 2 2                                15.6               14             7.16
## 3 3 (Jan 6 - Mar 29)               17.0               15.7           7.41
## 4 3 (Oct 28 - Dec 21)              17.2               15.7           7.52
## 5 3.1                              16.9               16.1           8.05
## 6 3.2                              18.5               17.4           9.07
## 7 3.3                              18.6               17.2           9.07
## 8 3.4                              19.2               18.3           9.23
```

Create a Bar plot to compare the mean utilization across the different phases.

```r
ggplot(phase_summary, aes(x = Phase, y = mean_utilization, fill = Phase)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Mental Health Care Utilization by Phase",
       x = "Phase",
       y = "Mean Utilization") +
  theme_minimal() + theme(axis.text.x = element_text(size = 7.5)) +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 10))  # Wrap labels
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).
```



Create a line plot to showing trend of utilization throught the phases.

```
ggplot(phase_summary, aes(x = as.factor(Phase), y = mean_utilization, group = 1)) +
  geom_line(color = "black", size = 1) +
  geom_smooth(method = "lm",se = FALSE, color = "blue") +
  geom_point(size = 3, color = "red") +
  labs(title = "Trend of Mental Health Care Utilization Across Phases",
       x = "Phase",
       y = "Mean Utilization") +
  theme_minimal() +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 10))  # Wrap labels
```

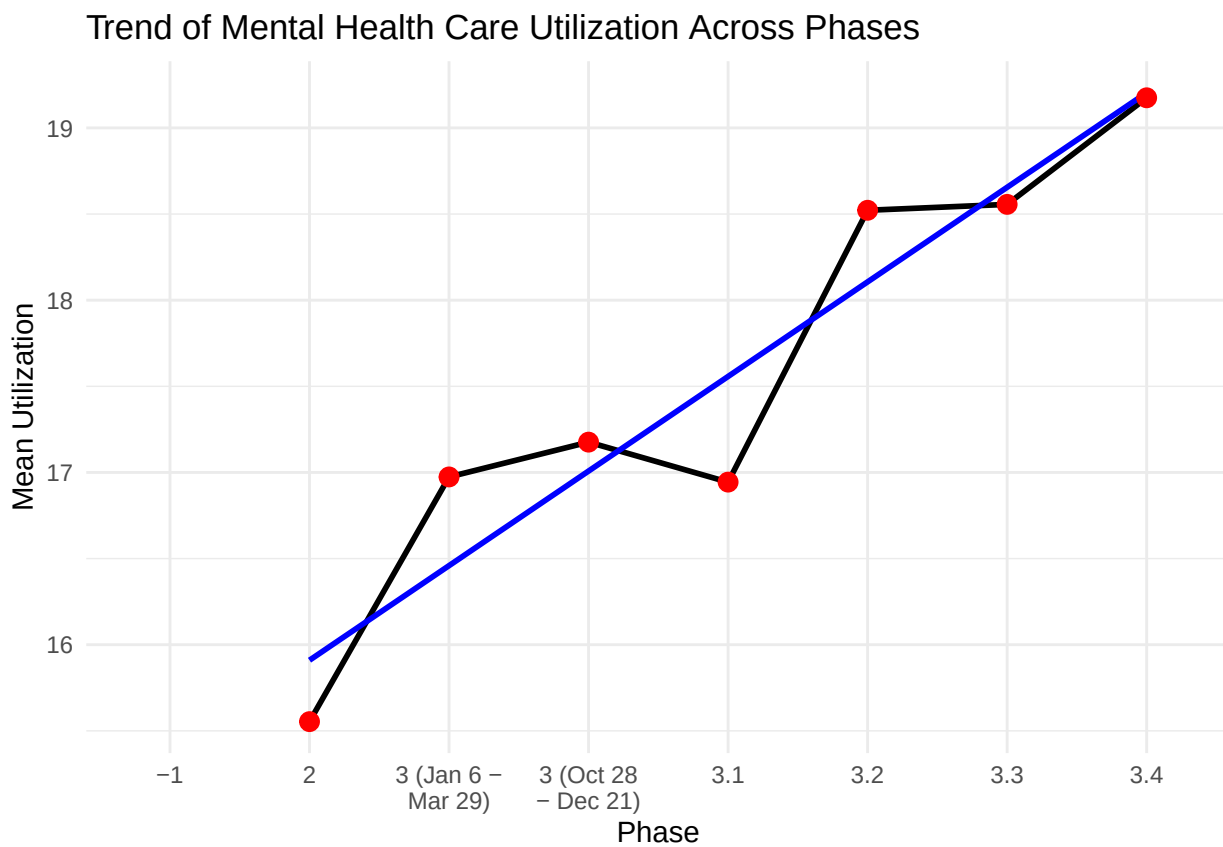
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

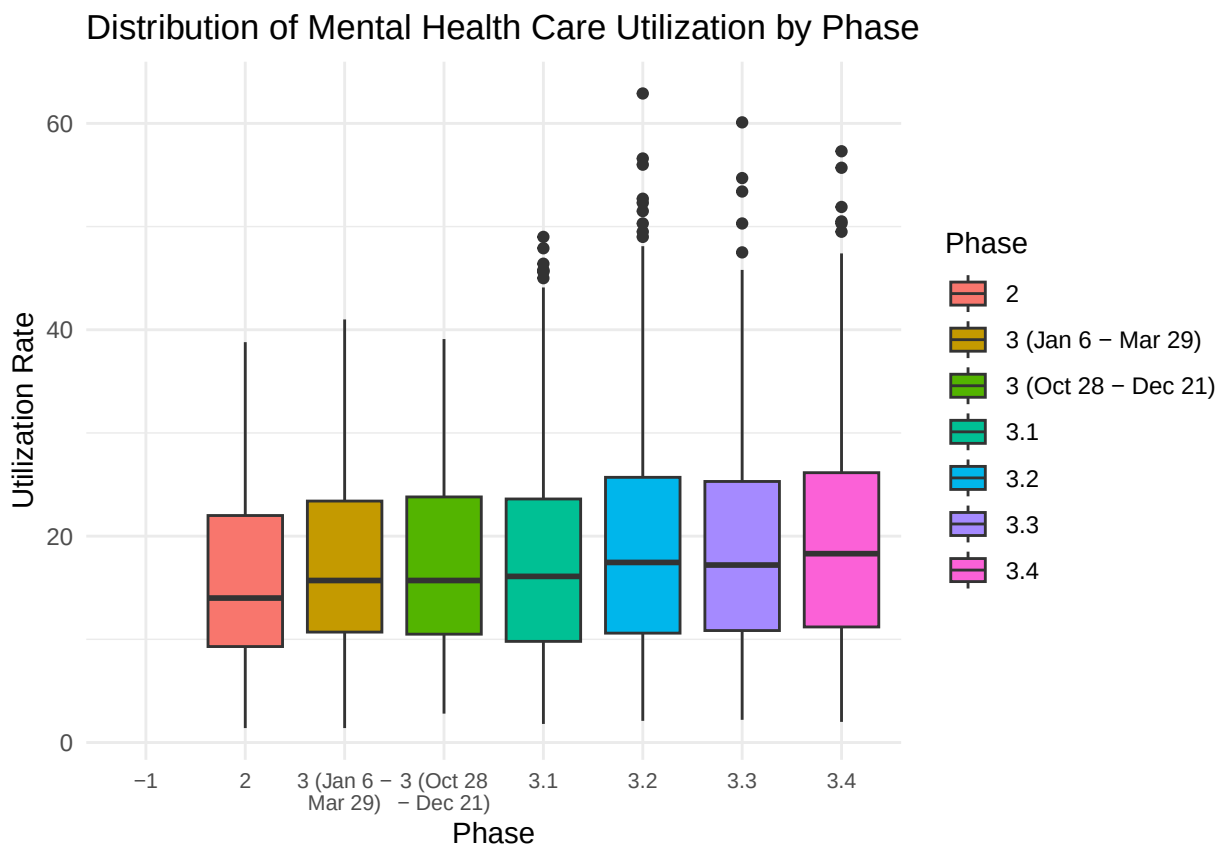
```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



Create a box plot to show full distribution of data points in each phase.

```r
ggplot(data, aes(x = as.factor(Phase), y = Value, fill = Phase)) +
  geom_boxplot() +
  labs(title = "Distribution of Mental Health Care Utilization by Phase",
       x = "Phase",
       y = "Utilization Rate") +
  theme_minimal() + theme(axis.text.x = element_text(size = 7.5)) +
  scale_x_discrete(labels = function(x) stringr::str_wrap(x, width = 10))  # Wrap labels
```

```
## Warning: Removed 490 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```r
# ================== REGRESSION ANALYSIS ================== #
# Linear Regression: Predicting mean_utilization using Phase
regression_model2 <- lm(mean_utilization ~ as.numeric(Phase), data = phase_summary)
```

```
## Warning in eval(predvars, data, env): NAs introduced by coercion
```

```r
regression_summary <- summary(regression_model2)
print(regression_summary)
```

```
##
## Call:
## lm(formula = mean_utilization ~ as.numeric(Phase), data = phase_summary)
##
## Residuals:
##        2        5        6        7        8
##  0.17426 -1.04331  0.29770  0.09483  0.47652
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        10.6376     1.8513   5.746   0.0105 *
## as.numeric(Phase)   2.3708     0.6084   3.897   0.0300 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6937 on 3 degrees of freedom
```

```
##   (3 observations deleted due to missingness)
## Multiple R-squared:  0.835,  Adjusted R-squared:   0.78
## F-statistic: 15.19 on 1 and 3 DF,  p-value: 0.02998
```

```r
# ================== CORRELATION ANALYSIS ================== #
# Spearman correlation since Phase is categorical (ordinal)
correlation_test2 <- cor.test(as.numeric(phase_summary$Phase), phase_summary$mean_utilization, method =
```

```
## Warning in cor.test(as.numeric(phase_summary$Phase),
## phase_summary$mean_utilization, : NAs introduced by coercion
```

```r
print(correlation_test2)
```

```
##
##  Spearman's rank correlation rho
##
## data:  as.numeric(phase_summary$Phase) and phase_summary$mean_utilization
## S = 4.4409e-15, p-value = 0.01667
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
##   1
```

```r
# ================== HYPOTHESIS TESTING ================== #
# Normality test (Shapiro-Wilk)
shapiro_test5 <- shapiro.test(phase_summary$mean_utilization)  # If p < 0.05, data is NOT normal
print(shapiro_test5)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  phase_summary$mean_utilization
## W = 0.93212, p-value = 0.5691
```

```r
anova_test2 <- aov(mean_utilization ~ as.factor(Phase), data = phase_summary)
summary(anova_test2)
```

```
##                  Df Sum Sq Mean Sq
## as.factor(Phase)  6  9.422    1.57
## 1 observation deleted due to missingness
```

Statistical Analysis of Data

Indicator Summary - The survey data reveals that 25.8% of adults reported utilizing mental health resources in the past four weeks, with 22.3% taking prescription medication, 11.2% needing but not receiving counseling/therapy, and 10.5% receiving counseling/therapy. Both the bar graph and box plot illustrate that individuals using prescription medication generally have higher utilization rates. The box plot also shows greater variability in utilization for prescription medication users compared to those seeking counseling. Additionally, the presence of outliers suggests that some individuals require significantly more mental health care than others. The bar graph further indicates that those who both took prescription medication and received counseling/therapy had the highest mean utilization, followed by those who only took prescription

medication, then those who needed but didn't receive counseling/therapy, and lastly those who only received counseling/therapy.

State Summary- Both the bar and error plot displays the average mental health care utilization across different states but doesn't establish a relationship between state and utilization. This suggests that further research is needed to explore potential factors influencing mental health care utilization, such as geographic location, climate, religion, or political beliefs. Statistical analysis confirms that state is not a significant predictor of mental health care utilization. This is supported by the regression analysis (p = 0.288) and Spearman's correlation (rho = 0.099, p = 0.482). The Shapiro-Wilk test (p = 0.04398) indicates that the mean utilization is not normally distributed, therefore a Kruskal-Wallis test (p = 0.4737) was performed. These findings suggest that the variable state is not a strong factor in predicting mental health care utilization.

Group Summary - The column plot visualizes that sexual orientation has the highest mean utilization among the different demographics, followed by gender identity, disability status, and the presence of symptoms of anxiety and depression. The Shapiro-Wilk test yielded a p-value of 0.02361, which is less than 0.05, indicating that the data is not normally distributed. Due to the non-normal distribution of the data, the nonparametric Kruskal-Wallis test was used instead of ANOVA, resulting in a p-value of 0.4373. This implies that there is no significant difference in mental health utilization across the different groups. The data also reveals that the variable national estimate displays a central tendency of utilization at a moderate rate with relatively lower variability. However, analysis By State shows high variability, suggesting significant differences in utilization rates among states. Further analysis By Sexual Orientation shows a wider spread, implying that different sexual orientation groups experience varied mental health care utilization. By Sex has a narrower spread compared to other groups, indicating more consistent utilization across male and female groups. By Race/Hispanic Ethnicity, data shows moderate variation in utilization, suggesting that some racial/ethnic groups have different access or utilization rates. Additionally, data By Presence of Anxiety/Depression Symptoms shows higher utilization, as expected, with a broad spread. By Gender Identity, data has a large spread, indicating diverse utilization patterns among different gender identities. By Education, a moderate spread suggests differences in mental health care access based on education level. By disability status, data shows higher utilization rates, possibly indicating greater mental health needs among individuals with disabilities. Lastly, data By Age shows variation, likely reflecting differences in mental health care needs and access across age groups. Overall, groups like sexual orientation, gender identity, and state have larger variability, suggesting disparities in mental health care utilization. Additionally, the presence of symptoms of anxiety or depression is associated with higher utilization. People with disabilities also show higher utilization rates, likely due to greater healthcare needs. Finally, state-level differences show a large variability across states, suggesting regional differences in access or availability of mental health care.

Age Summary- The data analysis reveals that mental health utilization decreases with age. This is supported by the Spearman's rank correlation, which shows a significant negative relationship between age and mental health utilization (rho = -0.4731, p = 0.011). This means that as age increases, mental health utilization significantly decreases. However, the regression analysis modeling this relationship was not statistically significant (p = 0.3734), and only explained 24.59% of the variance (R-squared value). Additionally, the Kruskal-Wallis test showed no significant difference in mental health utilization among different age groups (p = 0.38). While the Spearman's correlation suggests a significant negative trend between age and mental health utilization, the overall statistical analysis indicates that this trend is not strong enough to be considered statistically significant.

Sex Summary- The data analysis reveals that females tend to utilize mental health services more than males. This observation is supported by the visual representation of the data in the plots. However, the statistical tests conducted do not provide strong evidence for a significant difference in mental health utilization between sexes.The regression analysis indicates that males have a lower average mental health utilization compared to females, but this difference is not statistically significant (p = 0.180). Additionally, the model only explains 27.77% of the variance in the data, suggesting that sex is not a strong predictor of mental health utilization. The Spearman correlation shows a moderate negative correlation (rho = -0.4364), indicating that males tend to have lower utilization. However, this relationship is not statistically significant (p = 0.2797). The t-test further confirms the lack of a statistically significant difference in mental health utilization between males and females (p = 0.192). The wide confidence interval associated with the t-test suggests uncertainty in the

estimated difference and highlights the variability in the data. Overall, the results suggest that while there may be a trend towards higher mental health utilization in females, sex is not a key predictor of mental health utilization.

Disability Summary- The results of the statistical analysis, including regression analysis, Spearman's rank correlation, and t-test hypothesis testing, all indicate that individuals with disabilities have a significantly higher utilization of mental health services compared to those without disabilities. The regression analysis demonstrates that disability status accounts for 50.53% of the variance in mental health utilization, and the difference between the two groups is statistically significant ($p = 0.048$). The Spearman's rank correlation further supports this finding, revealing a strong negative correlation between disability status and mental health utilization (rho = -0.7638, $p = 0.0274$). While the t-test hypothesis testing yielded a p-value slightly above 0.05 ($p = 0.06196$), the trend still suggests a meaningful difference in mental health utilization between individuals with and without disabilities. Although the confidence interval includes 0, indicating a possibility of no effect, the effect size is large, suggesting a substantial difference between the two groups. These findings are also visually supported by the plots, which illustrate a higher utilization of mental health services among individuals with disabilities.

Education Summary - The bar graph illustrates that individuals with some college or an associate's degree utilized mental health services the most, followed by those with a bachelor's degree or higher. Those with less than a high school diploma were next, with those with a high school diploma or GED using mental health services the least. Statistical analysis emphasizes that there may not be a relationship between education level and mental health service utilization, as the results appear random or influenced by an unknown factor. The p-value of 0.6874 and Spearman's rho of 0.1091 indicate that education level does not have a strong or significant correlation with mental health utilization. Additionally, it is not a good predictor of mental health utilization, and there are no significant differences between the education levels. With an R square of 0.0789, only 7.89% of the variance is explained. The p-value of 0.7951 is also not significant. The Shapiro test, with a p-value of 0.1331, confirms that the data follows a normal distribution, as the p-value is greater than 0.05. The Kruskal-Wallis hypothesis test, with a p-value of 0.5011 (much greater than 0.05), also confirms that there is no significant difference in mental health utilization by education level.

Gender Identity Summary - The bar plot and statistical analysis show that gender identity significantly impacts mental health utilization. Transgender individuals have the highest utilization, followed by cis-gender females, and then cis-gender males. The data follows a normal distribution (Shapiro-Wilk normality test, $W = 0.9272$, $p = 0.3513$). The Kruskal-Wallis test (p-value = 0.0183) confirms a statistically significant difference in mental health utilization between gender groups. Spearman's rank correlation analysis indicates a moderate and significant correlation between gender and mental health care utilization (Spearman's rho = 0.5913, p-value = 0.0429). The strong model ($R^2 = 0.7842$) indicates that 78.42% of the variance is explained by gender identity. Transgender individuals have significantly higher utilization than cis-gender females, but there is no significant difference in utilization between cis-gender males and females.

Race Summary - The bar plot examining average mental health service utilization by race and ethnicity reveals that "Non-Hispanic, other races and multiple race" individuals have the highest utilization. This is followed by "Non-Hispanic White, single race" individuals, then Hispanic, "Non-Hispanic Black, single race", and lastly "Non-Hispanic Asian, single race" individuals. A scatter plot provides a more detailed breakdown by specific indicators. Statistical analysis indicates significant differences in mental health care utilization across racial and ethnic groups. Specifically, the "Non-Hispanic, other races and multiple race" group exhibits significantly higher utilization compared to the "Non-Hispanic Asian, single race" group. However, the overall regression model only explains a low amount of variance in utilization (38.52%), as shown by the R-squared value of 0.3852. This suggests that race alone is not a strong predictor of mental health care utilization. While the Spearman's rho = 0.4599 suggests a moderate positive correlation between race and utilization, and this correlation is statistically significant (p-value = 0.0414), the Kruskal-Wallis test (a non-parametric alternative to ANOVA) indicates that race does not have a significant impact on mental health utilization (Chi-squared = 7.8143, df = 4, $p = 0.0986$). This is further supported by the normality test ($p > 0.05$), which shows that the data follows a normal distribution.

Sexual Orientation Summary - The data analysis reveals that sexual orientation is a significant factor in mental health service utilization. Bisexual individuals have the highest average utilization, followed by gay

or lesbian individuals, and then straight individuals. This trend is further supported by a scatter plot that breaks down utilization by specific indicators and a statistical summary that shows a significant difference in utilization across sexual orientation groups, with the "Straight" group having significantly lower utilization compared to the "Bisexual" group. The overall model, with an R-squared value of 0.538, explains a moderate amount of variance in utilization. The normality of the data, as indicated by a p-value of 0.0308, supports the validity of the ANOVA and linear regression results.

Presence of Symptoms - The statistical analysis revealed that individuals who reported experiencing symptoms of anxiety or depression had a significantly higher average utilization of mental health care services compared to those who did not. This result is statistically significant (* indicates $p < 0.05$). The R^2 value of 0.6575 indicates that approximately 65.75% of the variance in Average Value can be explained by the presence of symptoms. The F-statistic (11.52, $p = 0.0146$) confirms the overall significance of the model. The Spearman correlation (rho = 0.8729, $p = 0.004659$) demonstrates a strong positive correlation between the presence of symptoms and average utilization value, suggesting a strong association even in cases where normality assumptions are violated. Additionally, the Welch two-sample t-test ($p = 0.01981$) further supports the existence of a significant difference between the two groups. The plots, regression analysis, correlation, and t-test all consistently confirm a significant positive relationship between mental health symptoms and the average utilization of mental health services.

Phase Summary - The results show a statistically significant positive correlation between phase and mean utilization, with mental health care utilization increasing as the phase increases. This trend is supported by the line plot with a positive regression line and the bar plot, both indicating that average utilization increases with phase. The regression model explains 83.5% of the variance in mental health utilization, with an adjusted R^2 of 0.78, showing that the model still explains 78% of the variance when adjusted for predictors. The overall model is statistically significant (F-statistic = 15.19, $p = 0.02998$). Spearman's rank correlation (p-value = 0.01667) further confirms the significant positive correlation between phase and mean utilization. The Shapiro-Wilk Normality Test (p-value = 0.569135) confirms data normality, but the subsequent ANOVA did not produce a p-value.